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一種多變量函數主成分分析的混合方法

A Hybrid Approach for Multivariate Functional Principal
Component Analysis

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摘要

近來，多變量函數數據愈來愈普遍。由主成分分析擴展而成的函數主成分分析 (FPCA) 是函數數據分析的基礎工具。儘管 FPCA 已經擴展到處理多變量函數數據，在處理多個函數中具有不同值域的稀疏觀測資料時仍存在一些問題。對於多變量函數主成分 (MFPCs) 的估計以及多變量函數主成分分數 (MFPC scores) 的預測，我們整合了先前研究的想法，提出了一個混合的方法。具體上，我們修改 [Happ and Greven \(2018\)](#) 中一個連結 FPCA 和多變量 FPCA 的矩陣之元素，利用 FPCA 中的特徵值估計作為對角元素，而非對角區塊內的元素由交叉共變異函數的傅立葉係數與個別單變量 FPCA 的估計特徵函數所組成。對於 MFPC scores 的預測，我們考慮了兩種方法；其一利用上述修改後的矩陣並遵循了 [Happ and Greven \(2018\)](#) 的流程，其二採用了 [Chiou et al. \(2014\)](#) 的加權最小平方法 (WLS) 的想法。我們所提出的方法在各個實驗設定下至少會與 [Happ and Greven \(2018\)](#) 和 [Chiou et al. \(2014\)](#) 兩者中的最佳方法一樣好。我們觀察到，以 WLS 估計 MFPC scores 在稀疏資料或是資料的各變數尺度差異大之情況下是適合的，而另一個提出的方法則對於各變數的觀測值數量較不敏感。

關鍵字：特徵分解、函數數據分析、函數主成分分析、多維度函數數據、多變量函數數據、加權最小平方法



Abstract

Data with multiple functional observations has become prevalent recently. Functional principal component analysis (FPCA), extended from principal component analysis, is a fundamental tool in functional data analysis. Although FPCA has been further extended to handle multivariate functional data, issues persist with sparse observations in multivariate functions over different domains. To estimate multivariate functional principal components (MFPCs) and to predict MFPC scores, we integrated the ideas in existing work and proposed a hybrid strategy. Specifically, we modified the elements in the matrix linking FPCA with multivariate FPCA in [Happ and Greven \(2018\)](#), and utilized the estimated eigenvalues in FPCA for the diagonal elements while the off-diagonal blocks consist of the Fourier coefficients of the estimated cross covariance functions represented with the estimated eigenfunctions of each univariate FPCA. To predict the MFPC scores, we considered two approaches; first, we followed [Happ and Greven \(2018\)](#) with the modified matrix, and second, we employed the weighted least squares (WLS) method by [Chiou](#)

et al. (2014). Our proposed approaches perform at least as the best between Chiou et al. (2014) and Happ and Greven (2018) under various simulation settings. Further, predicting the MFPC scores with the WLS method is more appropriate for sparse data or data with significant different scales between variables, while the other proposed method is less sensitive to the data where the scales of number of observations vary with variables.

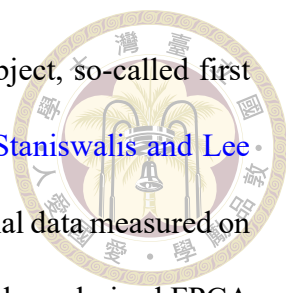
Keywords: eigendecomposition, functional data analysis, functional principal component analysis, multidimensional functional data, multivariate functional data, weighted least squares



Chapter 1 Introduction

Functional data is a specialized form of data, where each sample comprises measurements over a continuous domain, such as time or space, and we may have interests in capturing complex patterns and dynamics in the data that evolve over time or space. Examples include electrocardiogram curves measured over time, meteorological data measured over space and time, and so forth. Developing statistical approaches to analyze these types of data has become the focus of research in modern statistics. In functional data analysis, time is treated as a continuous domain, and data is viewed as a mapping rather than individual data points ([Ramsay, 1982](#)). However, functional data can only be observed at discrete points in practice, and thus it is necessary to transform these sample points into a functional form through some smoothing methods. Compared to multivariate analysis, functional data analysis can handle irregular observations or missing data more effectively and provides smoother estimates.

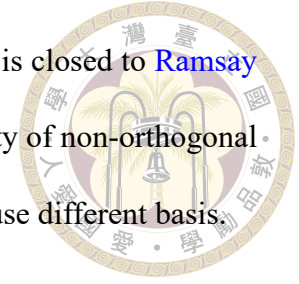
Functional data is intrinsically of infinite dimension, which not only brings rich information but challenges to both theoretical and numerical investigations. Analogous to principal component analysis in multivariate analysis, functional principal component analysis (FPCA) is capable of extracting the primary directions of data variation and represents the infinite-dimensional process of the original data efficiently. We can apply functional principal components (FPCs) to describe the variation modes of such curves. Existing FPCA



methods predominantly focus on the cases with one function per subject, so-called first generation functional data ([Wang et al., 2016](#)). To name but a few, [Staniswalis and Lee \(1998\)](#) introduced kernel-based FPCA, applying to univariate functional data measured on irregularly spaced time points; [Kayano and Konishi \(2009\)](#) introduced regularized FPCA using Gaussian basis functions, simplifying the computation of cross-product matrices; [Huang et al. \(2009\)](#) extended one-way FPCA to a two-way method by regularizing the left and right singular vectors obtained through singular value decomposition; [Allen \(2013\)](#) adapted FPCA methods for tensor data; [Song and Li \(2021\)](#) introduced non-linear relationships in functional data analysis using two separable Hilbert spaces, incorporating additivity to avoid the curse of dimensionality and extending FPCA to non-linear functions.

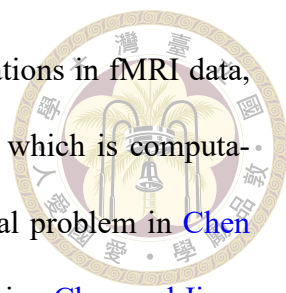
Data collected in the type of multivariate functional data becomes much more common recently. [Dai and Genton \(2018\)](#) analyzed the game controller gestures data, which involves changes in coordinates on three spatial axes (X, Y, Z) over a period of time. A well-known gait cycle data was studied by [Ramsay and Silverman \(2005\)](#), which examined the variation of hip and knee joint angles during the gait cycle. They performed FPCA independently on each variable, and found significant cross-correlation between the estimated FPC scores of the two variables. The independent analysis made the results difficult to interpret. Performing FPCA on correlated variables simultaneously seems more reasonable. For the analysis of multivariate functional data, we need to consider not only the stochastic process nature of functional data but also handle the cross-correlation between variables. [Berrendero et al. \(2011\)](#) performed PCA on the empirical cross-covariance matrix of multivariate functional observations made at regular grids and represented FPCs by curves through interpolation, while [Ramsay and Silverman \(2005\)](#) summarized the

principal components by scalar. [Jacques and Preda \(2014\)](#)'s method is closed to [Ramsay and Silverman \(2005\)](#), while they additionally explored the possibility of non-orthogonal basis, allowing each dimension of multivariate functional curves to use different basis.



Functional data can be irregular, which means the observation time points may vary for each sample, and the number of observations may also be different. The issue of sparseness may occur when most samples have few observations. To properly perform FPCA on sparse data, existing methods mainly focus on mixed effects with smoothing techniques, e.g., [Staniswalis and Lee \(1998\)](#); [James et al. \(2000\)](#); [Rice and Wu \(2001\)](#); [Yao et al. \(2005\)](#). The above methods are all built upon univariate functional data, and currently, the methods addressing the sparsity issues in multivariate functional data remain relatively limited. [Chiou et al. \(2014\)](#) introduced a normalized approach for multivariate FPCA (MFPCA) based on the normalized covariance operator, calculating FPC scores using weighted least squares. [Happ and Greven \(2018\)](#) proposed a framework that bridges univariate FPCA and multivariate FPCA through their Karhunen-Loève expansions and the corresponding FPC scores. [Li et al. \(2020\)](#) proposed a fast estimation method for calculating the covariance function, utilizing tensor-product B-splines to enable the eigendecomposition of the covariance operator.

Existing FPCA-related studies typically consider one-dimensional (1D) functional data. However, real-world data often involves multi-dimensional data, such as studies using brain imaging to identify disease characteristics ([Shi et al., 2024](#)) and meteorological features of neighboring regions ([Shen et al., 2023](#)). If data is multi-dimensional, 1D FPCA approaches using local smoothing or basis expansion may suffer from the curse of dimensionality ([Consagra et al., 2024](#)) and lose spatial information if vectorizing the data. Nonetheless, there have been numerous studies on multi-dimensional FPCA. [Chen](#)



[et al. \(2015\)](#) proposed 3D Image FPCA to analyze brain signal variations in fMRI data, using a smoothing method to estimate the 6D covariance function, which is computationally intensive. [Li et al. \(2019\)](#) addressed the large computational problem in [Chen et al. \(2015\)](#) with singular value decomposition and penalized smoothing. [Chen and Jiang \(2017\)](#) used local linear smoothing to estimate the multi-dimensional covariance function and addressed computational issues with fast Fourier transform strategy and parallel computation. [Di Salvo et al. \(2015\)](#) proposed a model that combines a generalized additive model with FPCA and applied it to multi-dimensional (spatio-temporal) multivariate data. Despite these advancements in multi-dimensional FPCA, research on multi-dimensional MFPCA remains scarce.

The method of [Happ and Greven \(2018\)](#) is not only applicable to sparse data but also suitable for functional observations made on different domains and multi-dimensional functional data. For sparse data, PACE ([Yao et al., 2005](#)) was employed to predict the FPC scores in [Happ and Greven \(2018\)](#). However, due to the conditional expectation structure, the estimated covariance based on the FPC scores predicted by PACE is wrongly estimated, motivating this study. The estimation of the covariance function is crucial for multivariate functional principal components (MFPCs) and thus the MFPC scores. We aim for MFPCA to be applicable to various data types such as sparse, multivariate, and multi-dimensional in addition to achieving precise and effective estimation. [Chiou et al. \(2014\)](#) proposed a normalized MFPCA to handle the cases where certain components of the multivariate functional data exhibit larger variability and predict the MFPC scores through weighted least squares, with the ratio of measurement error variance to variance function as weights. This method performs well for sparse multivariate functional data across different variable scales. The numerical experiments in [Chen \(2023\)](#) indicate that

the increase in estimation error is closely related to the increase in variability between variables, data sparsity, and measurement errors. Our research aims to combine the advantages of [Happ and Greven \(2018\)](#) and [Chiou et al. \(2014\)](#) to propose a hybrid MFPCA approach, making it suitable for various configurations in multivariate functional data.

The rest of this thesis is organized as follows. Chapter 2 introduces functional data, FPCA, MFPCA, and other related methods along with the notations and required assumptions. Chapter 3 explains our two proposed methods, detailing how we combine elements from past studies in our proposed approaches. Chapter 4 contains the simulation results with discussions. The proposed methods are applied to the medical data from a longitudinal retrospective study in chapter 5, while a comprehensive summary and subsequent discussion of the entire study are provided in the last chapter.



Chapter 2 Multivariate Functional Principal Component Analysis

We first introduce univariate functional data ($p = 1$ for multivariate functional data). Let $X : \mathcal{T} \rightarrow \mathbb{R}$ be a random function in the Hilbert space $L^2(\mathcal{T})$, which denotes the space of square-integrable functions on the compact set $\mathcal{T}$, with $f : \mathcal{T} \rightarrow \mathbb{R}$, $(\int_{\mathcal{T}} f^2)^{1/2} < \infty$. The inner product in $L^2(\mathcal{T})$ is denoted as $\langle f, g \rangle_2 = \int_{\mathcal{T}} f(t)g(t)dt$, for $f, g \in L^2(\mathcal{T})$, with the norm $\|\cdot\|_2 = \langle \cdot, \cdot \rangle_2^{1/2}$. The mean function and covariance function are defined as

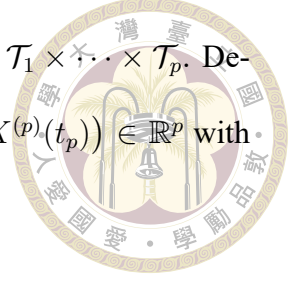
$$\mu(t) = \mathbb{E}(X(t)), \text{ and } G(t, u) = \text{Cov}(X(t), X(u)), \text{ for } t, u \in \mathcal{T},$$

respectively, and the variance function is defined as $V(t) := G(t, t)$.

2.1 Multivariate Functional Data

Each sample of multivariate functional data consists of p functions $X = (X^{(1)}, \dots, X^{(p)})$, with the k th variable $X^{(k)} : \mathcal{T}_k \rightarrow \mathbb{R}$, assumed to be in $\mathcal{H} := L^2(\mathcal{T}_1) \times \dots \times L^2(\mathcal{T}_p)$, a union of Hilbert spaces $L^2(\mathcal{T}_k)$ with respect to (w.r.t.) the domain $\mathcal{T}_k$ and $\mathcal{T}_k$ is a compact

set in $\mathbb{R}^{d_k}$, where d_k are positive integers. For convenience, let $\mathcal{T} = \mathcal{T}_1 \times \cdots \times \mathcal{T}_p$. Denote the measurement of X at $\mathbf{t}$ as a vector $X(\mathbf{t}) = (X^{(1)}(t_1), \dots, X^{(p)}(t_p)) \in \mathbb{R}^p$ with $\mathbf{t} := (t_1, \dots, t_p) \in \mathcal{T}$.



The mean function and the covariance function of $X(\mathbf{t})$ are, respectively, defined as

$$\mu(\mathbf{t}) := \mathbb{E}(X(\mathbf{t})) = (\mathbb{E}[X^{(1)}(t_1)], \dots, \mathbb{E}[X^{(p)}(t_p)]), \quad \forall \mathbf{t} \in \mathcal{T},$$

and

$$G(\mathbf{t}, \mathbf{u}) := \mathbb{E}([X(\mathbf{t}) - \mu(\mathbf{t})]^\top \otimes [X(\mathbf{u}) - \mu(\mathbf{u})]),$$

with the (k, l) -th block

$$G_{kl}(t_k, u_l) := \text{Cov}(X^{(k)}(t_k), X^{(l)}(u_l)), \quad \forall t_k \in \mathcal{T}_k, u_l \in \mathcal{T}_l.$$

Also, the variance function is $V_k(t_k) := G_{kk}(t_k, t_k)$.

Following [Happ and Greven \(2018\)](#), the inner product in $\mathcal{H}$ is defined as

$$\langle \mathbf{f}, \mathbf{g} \rangle_{\mathcal{H}} := \sum_{k=1}^p \langle f^{(k)}, g^{(k)} \rangle_2 = \sum_{k=1}^p \int_{\mathcal{T}_k} f^{(k)}(t_k) g^{(k)}(t_k) dt_k, \quad (2.1)$$

with the norm $\|\cdot\|_{\mathcal{H}} = \langle \cdot, \cdot \rangle_{\mathcal{H}}^{1/2}$, where $\mathbf{f}, \mathbf{g} \in \mathcal{H}$, $\mathbf{f} = (f^{(1)}, \dots, f^{(p)})$ and $\mathbf{g} = (g^{(1)}, \dots, g^{(p)})$ with $f^{(k)}, g^{(k)} \in L^2(\mathcal{T}_k)$. The inner product (2.1) could be generalized to include weights, i.e., $\langle \mathbf{f}, \mathbf{g} \rangle_{\mathcal{H}} := \sum_{k=1}^p w_k \langle f^{(k)}, g^{(k)} \rangle_2$. [Happ and Greven \(2018\)](#) suggested to consider weights for the cases where the variable domains or the variable variations are significantly different, but it could not tackle the issues due to data sparseness.

2.2 FPCA



Before introducing multivariate FPCA, we briefly review FPCA. Let us consider a random function $X^{(k)}$ for $k \in \{1, \dots, p\}$, with mean function $\mu^{(k)}(t_k)$ and covariance function $G_{kk}(t_k, u_k)$ for $t_k, u_k \in \mathcal{T}_k$. For $f^{(k)} \in L^2(\mathcal{T}_k)$, the function Γ^k induces the covariance operator

$$(\Gamma^k f^{(k)})(t_k) := \int_{\mathcal{T}_k} G_{kk}(t_k, u_k) f^{(k)}(u_k) du_k = \langle G_{k \cdot}(t_k, \cdot), f^{(k)} \rangle_2, \quad \forall t_k \in \mathcal{T}_k.$$

Further, it can be shown that if $\mathbb{E}(\|X^{(k)}\|_2^2) < \infty$, then Γ^k is a symmetric, positive Hilbert-Schmidt operator in $L^2(\mathcal{T}_k)$ (Kokoszka, 2012). By Mercer's Theorem (Mercer, 1909),

$$G_{kk}(t_k, u_k) = \sum_{m=1}^{\infty} \lambda_m^{(k)} \phi_m^{(k)}(t_k) \phi_m^{(k)}(u_k),$$

where $\lambda_1^{(k)} \geq \lambda_2^{(k)} \geq \dots \geq 0$ are the eigenvalues of Γ^k , and $\phi_m^{(k)}$'s are the corresponding orthogonal eigenfunctions, also known as the FPCs. The above eigenvalues and eigenfunctions are the solutions to the eigenequation

$$\Gamma^k \phi_m^{(k)} = \lambda_m^{(k)} \phi_m^{(k)}, \quad \forall m \in \mathbb{N}. \quad (2.2)$$

By Karhunen-Loève decomposition, the stochastic process $X^{(k)}$ can be represented as

$$X^{(k)}(t_k) = \mu^{(k)}(t_k) + \sum_{m=1}^{\infty} \xi_m^{(k)} \phi_m^{(k)}(t_k), \quad \forall t_k \in \mathcal{T}_k, \quad (2.3)$$

where

$$\xi_m^{(k)} = \langle X^{(k)} - \mu^{(k)}, \phi_m^{(k)} \rangle_2 \quad (2.4)$$

are the FPC scores, which satisfy

$$\mathbb{E}(\xi_m^{(k)}) = 0, \quad \mathbb{E}(\xi_m^{(k)} \xi_r^{(k)}) = \lambda_m^{(k)} I(m = r). \quad (2.5)$$



If we work on the normalized data, i.e., $Z^{(k)}(t_k) = V_k^{-1/2}(t_k)\{X^{(k)}(t_k) - \mu^{(k)}(t_k)\}$, and let $\tilde{\Gamma}^k$ be the covariance operator corresponding to $\text{Cov}(Z^{(k)}(t_k), Z^{(k)}(u_k))$, then $X^{(k)}(t_k)$ can be represented as

$$X^{(k)}(t_k) = \mu^{(k)}(t_k) + \sum_{m=1}^{\infty} \tilde{\xi}_m^{(k)} \left(V_k^{1/2} \tilde{\phi}_m^{(k)} \right) (t_k), \quad \forall t_k \in \mathcal{T}_k,$$

where the FPC scores $\tilde{\xi}_m^{(k)} = \langle Z^{(k)}, \tilde{\phi}_m^{(k)} \rangle_2$ satisfy $\mathbb{E}(\tilde{\xi}_m^{(k)}) = 0$ and $\mathbb{E}(\tilde{\xi}_m^{(k)} \tilde{\xi}_r^{(k)}) = \tilde{\lambda}_m^{(k)} I(m = r)$. The eigenvalues $\tilde{\lambda}_m^{(k)}$ and the corresponding eigenfunctions $\tilde{\phi}_m^{(k)}$ are the solutions to the eigenequation $\tilde{\Gamma}^k \tilde{\phi}_m^{(k)} = \tilde{\lambda}_m^{(k)} \tilde{\phi}_m^{(k)}, \quad \forall m \in \mathbb{N}$.

In practice, functional data are collected at discrete points and possibly contaminated with measurement errors. Let $x_{i,j}^{(k)}$ be the j th observation of $X^{(k)}$ of the i th sample, and $\mathbf{x}_i^{(k)} = \left(x_{i,1}^{(k)}, \dots, x_{i,n_i^{(k)}}^{(k)} \right)^\top$ be the $n_i^{(k)}$ observations of $X^{(k)}$ of the i th sample. By (2.3),

$$\begin{aligned} x_{i,j}^{(k)} &= \mu^{(k)}(t_{i,j}^{(k)}) + \sum_{m=1}^{\infty} \xi_{i,m}^{(k)} \phi_m^{(k)}(t_{i,j}^{(k)}) + \varepsilon_{i,j}^{(k)} \\ &\approx \mu^{(k)}(t_{i,j}^{(k)}) + \sum_{m=1}^{M^{(k)}} \xi_{i,m}^{(k)} \phi_m^{(k)}(t_{i,j}^{(k)}) + \varepsilon_{i,j}^{(k)}, \quad \forall i = 1, \dots, N, \quad j = 1, \dots, n_i^{(k)}, \end{aligned} \quad (2.6)$$

where $\varepsilon_{i,j}^{(k)}$ are assumed to be independently and identically distributed (*i.i.d.*) measurement errors with $\mathbb{E}(\varepsilon_{i,j}^{(k)}) = 0$, and $\text{Var}(\varepsilon_{i,j}^{(k)}) = \sigma_k^2$, and $M^{(k)}$ denotes certain truncation lag. We can choose $M^{(k)}$ by the fraction of variance explained (FVE) criterion, the formula of which will be provided in Section 3.1.

To estimate mean function, we apply a local linear smoother with bandwidth $h_\mu^{(k)}$ to

the observations; specifically, we solve the following minimization problem

$$\left(\hat{\beta}_0^{(k)}, \hat{\beta}_1^{(k)}\right) = \arg \min_{(\beta_0, \beta_1)} \sum_{i=1}^N \sum_{j=1}^{n_i^{(k)}} \left\{ x_{i,j}^{(k)} - \beta_0 - \beta_1(t_k - t_{i,j}^{(k)}) \right\}^2 K_1 \left(\frac{t_{i,j}^{(k)} - t_k}{h_{\mu}^{(k)}} \right), \quad (2.7)$$

where $K_1 : \mathbb{R} \rightarrow \mathbb{R}$ is a symmetric kernel with support $[-1, 1]$ and satisfying $\int K_1^2(u) du < \infty$. The estimated mean function is thus $\hat{\mu}^{(k)}(t_k) = \hat{\beta}_0^{(k)}(t_k)$.

To estimate the eigenfunctions, the eigenvalues and the FPC scores, we need to estimate the covariance function first and we apply a two-dimensional local linear smoother with bandwidth $h_{G_{kk}}$. Similar to K_1 , we let $K_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a symmetric kernel with support $[-1, 1] \times [-1, 1]$ and satisfying $\int \int K_2^2(u, v) du dv < \infty$. Also, let $\tilde{G}_{kk} \left(t_{i,j}^{(k)}, t_{i,j'}^{(k)} \right) = \left\{ x_{i,j}^{(k)} - \hat{\mu}^{(k)}(t_{i,j}^{(k)}) \right\} \left\{ x_{i,j'}^{(k)} - \hat{\mu}^{(k)}(t_{i,j'}^{(k)}) \right\}$ be a raw estimate of covariance surface with $\hat{\mu}^{(k)}(t_k)$ obtained by (2.7). Again, we solve the following problem to estimate G_{kk}

$$\left(\hat{\gamma}_0^{(k)}, \hat{\gamma}_1^{(k)}, \hat{\gamma}_2^{(k)}\right) = \arg \min_{(\gamma_0^{(k)}, \gamma_1^{(k)}, \gamma_2^{(k)})} \sum_{i=1}^N \sum_{1 \leq j \neq j' \leq n_i^{(k)}} \left\{ f \left(\gamma, (t_k, u_k), (t_{i,j}^{(k)}, t_{i,j'}^{(k)}) \right) \right\}^2 K_2 \left(\frac{t_{i,j}^{(k)} - t_k}{h_{G_{kk}}}, \frac{t_{i,j'}^{(k)} - u_k}{h_{G_{kk}}} \right), \quad (2.8)$$

where $f \left(\gamma, (t_k, u_k), (t_{i,j}^{(k)}, t_{i,j'}^{(k)}) \right) = \tilde{G}_{kk} \left(t_{i,j}^{(k)}, t_{i,j'}^{(k)} \right) - \gamma_0 - \gamma_1(t_k - t_{i,j}^{(k)}) - \gamma_2(u_k - t_{i,j'}^{(k)})$; the estimated covariance function is thus $\hat{G}_{kk}(t_k, u_k) = \hat{\gamma}_0^{(k)}(t_k, u_k)$. Now, we can estimate the eigenvalues $\lambda_m^{(k)}$ and eigenfunctions $\phi_m^{(k)}$ through the eigenequation (2.2), i.e.,

$$\int_{\mathcal{T}_k} \hat{G}_{kk}(t_k, u_k) \hat{\phi}_m^{(k)}(t_k) dt_k = \hat{\lambda}_m^{(k)} \hat{\phi}_m^{(k)}(u_k). \quad (2.9)$$

Since the data collected may be contaminated with measurement errors $\varepsilon_{i,j}^{(k)}$,

$$\text{Cov} \left(x_{i,j}^{(k)}, x_{i,l}^{(k)} \right) = G_{kk}(t_{i,j}, t_{i,l}) + \sigma_k^2 I(j = l),$$



where $\sigma_k^2 = \text{Var} \left(\varepsilon_{i,j}^{(k)} \right)$. Due to the existence of measurement errors, the estimator of G_{kk} becomes inconsistent if the cases $j = l$ are considered in (2.8).

Let $V^{(k)}(t_k) = G_{kk}(t_k, t_k) + \sigma_k^2$, and we could estimate $V^{(k)}(t_k)$ by applying a local linear smoother on $\left\{ \left(\tilde{G}_{kk}(t_{i,j}^{(k)}, t_{i,j}^{(k)}), t_{i,j}^{(k)} \right) \right\}_{i,j=1}^{N, n_i}$. Given that $G_{kk}(t_k, t_k)$ has been estimated by (2.8), the difference between $\hat{G}_{kk}(t_k, t_k)$ and $\hat{V}^{(k)}(t_k)$ can be used as an estimate of the measurement error variance. Specifically,

$$\hat{\sigma}_k^2 = \frac{1}{|\mathcal{T}'_k|} \int_{\mathcal{T}'_k} \{ \hat{V}^{(k)}(t_k) - \hat{G}_{kk}(t_k, t_k) \} dt_k, \quad (2.10)$$

where $|\mathcal{T}'_k|$ denote the length of $\mathcal{T}'_k$, and a reasonable choice of $\mathcal{T}'_k$ is

$$\mathcal{T}'_k = [\inf \{x : x \in \mathcal{T}_k\} + |\mathcal{T}_k|/4, \sup \{x : x \in \mathcal{T}_k\} - |\mathcal{T}_k|/4].$$

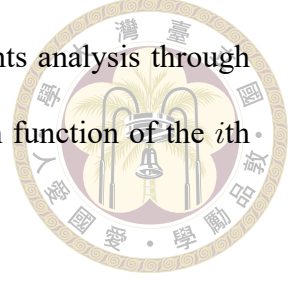
With the estimated mean functions and eigenfunctions, the FPC scores can be predicted by numerically integral approximation, i.e.,

$$\hat{\xi}_{i,m}^{(k)} = \left\langle \mathbf{x}_i^{(k)} - \hat{\boldsymbol{\mu}}_i^{(k)}, \hat{\Phi}_{i,m}^{(k)} \right\rangle_2 \approx \sum_{j=2}^{n_i^{(k)}} \left(t_{i,j}^{(k)} - t_{i,j-1}^{(k)} \right) \left(x_i^{(k)}(\zeta_{i,j}^{(k)}) - \hat{\mu}_i^{(k)}(\zeta_{i,j}^{(k)}) \right) \hat{\phi}_m^{(k)}(\zeta_{i,j}^{(k)}),$$

where $\hat{\boldsymbol{\mu}}_i^{(k)} = \left(\hat{\mu}_{i,1}^{(k)}, \dots, \hat{\mu}_{i,n_i^{(k)}}^{(k)} \right)^\top$, $\hat{\Phi}_{i,m}^{(k)} = \left(\hat{\phi}_m^{(k)}(t_{i,1}^{(k)}), \dots, \hat{\phi}_m^{(k)}(t_{i,n_i^{(k)}}^{(k)}) \right)^\top$, and $\zeta_{i,j}^{(k)} = \frac{1}{2} \left(t_{i,j}^{(k)} + t_{i,j-1}^{(k)} \right)$. However, integral approximation may not be appropriate for sparse functional data.

To predict the FPC scores for sparse functional data, [Yao et al. \(2005\)](#) proposed a

conditional expectation approach named PACE (principal components analysis through conditional expectation). Specifically, the m th FPC score of the k th function of the i th subject is given by



$$\hat{\xi}_{i,m}^{(k)} = \hat{\mathbb{E}} \left(\xi_{i,m}^{(k)} \mid \mathbf{x}_i^{(k)} \right) = \hat{\lambda}_m^{(k)} \hat{\Phi}_{i,m}^{(k)\top} \hat{\Sigma}_i^{(k)-1} \left(\mathbf{x}_i^{(k)} - \hat{\boldsymbol{\mu}}_i^{(k)} \right), \quad (2.11)$$

with the (r, s) entry of $\hat{\Sigma}_i^{(k)} \in \mathbb{R}^{n_i^{(k)} \times n_i^{(k)}}$ being $\hat{G}_{kk}(t_{i,r}, t_{i,s}) + \hat{\sigma}_k^2 I(r = s)$. As [Yao et al. \(2005\)](#) pointed out that $\hat{\xi}_{i,m}^{(k)}$ (2.11) is the best linear unbiased prediction of $\xi_{i,m}^{(k)}$, whether the Gaussian assumption holds or not.

2.3 Multivariate FPCA

For a p -variate function $\mathbf{f} = (f^{(1)}, \dots, f^{(p)})$, the function Γ induces the covariance operator $L^2(\mathcal{T}) \rightarrow L^2(\mathcal{T})$, $\mathbf{f} \rightarrow \Gamma \mathbf{f}$ as

$$(\Gamma \mathbf{f})^{[k]}(t_k) := \sum_{l=1}^p \int_{\mathcal{T}_l} G_{kl}(t_k, u_l) f^{(l)}(u_l) \mathrm{d}u_l = \langle G_{k\cdot}(t_k, \cdot), \mathbf{f} \rangle_{\mathcal{H}}, \quad \forall t_k \in \mathcal{T}_k, \quad (2.12)$$

where $G_{kl}(t_k, u_l)$ is the cross-covariance function of $X^{(k)}$ and $X^{(l)}$. In the following, $[k]$ denotes the k th block corresponding to the k th variable $X^{(k)}$. By Proposition 3 in [Happ and Greven \(2018\)](#),

$$G_{kk}(t_k, u_k) = \sum_{m=1}^{\infty} \omega_m \psi_m^{[k]}(t_k) \psi_m^{[k]}(u_k),$$

where $\omega_1 \geq \omega_2 \geq \dots \geq 0$ are the eigenvalues of Γ , ψ_m 's are the corresponding eigenfunctions, or called multivariate FPCs (MFPCs), and $\psi_m^{[k]}$ denotes the k th block of ψ_m corresponding to $X^{(k)}$. Notice that in the MFPCA, every variable shares the same eigenvalues ω_m . The eigenvalues and the corresponding eigenfunctions are the solutions to the

eigenequation

$$\Gamma \psi_m = \omega_m \psi_m, \quad \forall m \in \mathbb{N}. \quad (2.13)$$



The estimating form is represented as

$$\sum_{k=1}^p \int_{\mathcal{T}_k} \hat{G}_{kl}(t_k, u_l) \hat{\psi}_m^{[k]}(t_k) dt_k = \hat{\omega}_m \hat{\psi}_m^{[l]}(u_l). \quad (2.14)$$

By the multivariate Karhunen–Loève theorem (Happ and Greven, 2018), the p -variate stochastic process X can be represented as

$$X(\mathbf{t}) = \mu(\mathbf{t}) + \sum_{m=1}^{\infty} \eta_m \psi_m(\mathbf{t}), \quad \forall \mathbf{t} \in \mathcal{T},$$

where the MFPC score $\eta_m = \langle (X - \mu), \psi_m \rangle_{\mathcal{H}} = \sum_{k=1}^p \left\langle (X^{(k)} - \mu^{(k)}), \psi_m^{[k]} \right\rangle_2$ with $\mathbb{E}(\eta_m) = 0$ and $\mathbb{E}(\eta_m \eta_r) = \omega_m I(m = r)$.

To work on the multivariate normalized data $Z(\mathbf{t}) = (Z^{(1)}(t_1), \dots, Z^{(p)}(t_p))$, we let $D = \text{diag}(V_1^{1/2}(t_1), \dots, V_p^{1/2}(t_p))$, and $\tilde{\Gamma}$ be the covariance operator satisfying

$$(\tilde{\Gamma} \mathbf{f})^{[k]}(t_k) := \sum_{l=1}^p \int_{\mathcal{T}_l} \text{Cov}(Z^{(k)}(t_k), Z^{(l)}(u_l)) f^{(l)}(u_l) du_l = \langle G_{k \cdot}(t_k, \cdot), \mathbf{f} \rangle_{\mathcal{H}}, \quad \forall t_k \in \mathcal{T}_k.$$

The stochastic process $X(\mathbf{t})$ can thus be represented as

$$X(\mathbf{t}) = \mu(\mathbf{t}) + \sum_{m=1}^{\infty} \tilde{\eta}_m (D \tilde{\psi}_m)(\mathbf{t}), \quad \forall \mathbf{t} \in \mathcal{T},$$

where the MFPC scores $\tilde{\eta}_m = \langle Z, \tilde{\psi}_m \rangle_{\mathcal{H}} = \sum_{k=1}^p \langle Z^{(k)}, \tilde{\psi}_m^{[k]} \rangle_2$ satisfy $\mathbb{E}(\tilde{\eta}_m) = 0$ and $\mathbb{E}(\tilde{\eta}_m \tilde{\eta}_r) = \tilde{\omega}_m I(m = r)$. The eigenvalues $\tilde{\omega}_m$ and the corresponding eigenfunctions $\tilde{\psi}_m$ are the solutions to the eigenequation $\tilde{\Gamma} \tilde{\psi}_m = \tilde{\omega}_m \tilde{\psi}_m, \quad \forall m \in \mathbb{N}.$

When reconstructing the observed samples, we choose a finite number M , and use

the first M MFPCs and the corresponding MFPC scores as an approximation of multivariate Karhunen-Loève expansions of X , a multivariate version of (2.6). We can choose M by FVE, the same criterion of choosing truncation lag $M^{(k)}$ in the Karhunen-Loève expansion.

Proposition 5 of Happ and Greven (2018) enables us to link the calculation of MFPCs and MFPC scores based with their univariate counterparts. Assume each variable $X^{(k)}$ is centered, i.e., $\mu^{(k)}(t_k) = 0$, for simplicity and could be represented with a finite number of eigenfunctions by the Karhunen-Loève representation, i.e., $X^{(k)} = \sum_{m=1}^{M^{(k)}} \xi_m^{(k)} \phi_m^{(k)}$, where the eigenequations $\phi_m^{(k)}$ satisfy $\Gamma^k \phi_m^{(k)} = \lambda_m^{(k)} \phi_m^{(k)}$ and $\{\lambda_m^{(k)}\}_{m=1}^{M^{(k)}}$ are non-increasing and positive. Define $M^+ := \sum_{k=1}^p M^{(k)} \geq M$, where M is a chosen truncation lag of MFPCA. By the Karhunen-Loève representation, $X^{[k]} = \sum_{m=1}^M \eta_m \psi_m^{[k]}$, where ψ_m are the eigenfunctions satisfy $\Gamma \psi_m^{[k]} = \omega_m \psi_m^{[k]}$. The matrix $\mathbf{A} \in \mathbb{R}^{M^+ \times M^+}$ consists of $p \times p$ blocks of matrices. Specifically, the (k, l) th block matrix $\mathbf{A}^{[kl]} \in \mathbb{R}^{M^{(k)} \times M^{(l)}}$ with the (m, r) th entry

$$A_{mr}^{[kl]} = \text{Cov}(\xi_m^{(k)}, \xi_r^{(l)}), \quad \forall m = 1, \dots, M^{(k)}, \quad r = 1, \dots, M^{(l)}, \quad k, l = 1, \dots, p. \quad (2.15)$$

The MFPCs can thus be written as

$$\psi_m^{[k]}(t_k) = \sum_{j=1}^{M^{(k)}} [\mathbf{a}_m]_j^{[k]} \phi_j^{(k)}(t_k), \quad \forall t_k \in \mathcal{T}_k, \quad m = 1, \dots, M, \quad (2.16)$$

where $[\mathbf{a}_m]_j^{[k]} \in \mathbb{R}^{M^{(k)}}$ represents the j th element in the k th block of the m th eigenvector $\mathbf{a}_m$. The MFPC scores can be expressed by

$$\eta_m = \sum_{k=1}^p \sum_{j=1}^{M^{(k)}} [\mathbf{a}_m]_j^{[k]} \xi_j^{(k)}, \quad \forall m = 1, \dots, M. \quad (2.17)$$



Chapter 3 Estimation of MFPCs and MFPC Scores

The matrix $\mathbf{A}$ (2.15) is the most crucial part of the method proposed in [Happ and Greven \(2018\)](#) since it links the Karhunen-Loève representation of each $X^{(k)}$ with that of X . [Happ and Greven \(2018\)](#) suggested to employ the predicted FPC scores to construct $\mathbf{A}$ regardless of the sampling schemes of each variable. In particular, when the functional data are sparse and irregularly observed, PACE is employed to predict the FPC scores. Given that PACE is an shrinkage estimator, $\mathbf{A}$ constructed with PACE predicted FPC scores tend to be biased. As a consequence, we propose a procedure to generate a consistent estimate.

From (2.3), (2.12), (2.13) and the supplement material of [Happ and Greven \(2018\)](#), we obtain the following equation

$$\begin{aligned} (\Gamma\psi)^{[k]}(t_k) &= \sum_{l=1}^p \int_{\mathcal{T}_l} \text{Cov}(X_c^{(k)}(t_k), X_c^{(l)}(u_l)) \psi^{[l]}(u_l) \mathrm{d}u_l \\ &= \sum_{l=1}^p \sum_{m=1}^{M^{(k)}} \sum_{r=1}^{M^{(l)}} \text{Cov}(\xi_m^{(k)}, \xi_r^{(l)}) \phi_m^{(k)}(t_k) \int_{\mathcal{T}_l} \phi_r^{(l)}(u_l) \psi^{[l]}(u_l) \mathrm{d}u_l, \end{aligned}$$

where $X_c^{(k)}(t_k) = X^{(k)}(t_k) - \mu^{(k)}(t_k)$, and $\text{Cov}(\xi_m^{(k)}, \xi_r^{(l)})$ is equal to $A_{mr}^{[kl]}$ from (2.15),

which is the term we will concentrate on. Observe that

$$\begin{aligned}\widehat{\text{Cov}}(\xi_m^{(k)}, \xi_r^{(l)}) &= \hat{\mathbb{E}} \left[\widehat{\text{Cov}}(\xi_m^{(k)}, \xi_r^{(l)} \mid \tilde{x}^{(k)}, \tilde{x}^{(l)}) \right] + \widehat{\text{Cov}} \left[\hat{\mathbb{E}}(\xi_m^{(k)} \mid \tilde{x}^{(k)}), \hat{\mathbb{E}}(\xi_r^{(l)} \mid \tilde{x}^{(l)}) \right] \\ &= \hat{\mathbb{E}} \left[\widehat{\text{Cov}}(\xi_m^{(k)}, \xi_r^{(l)} \mid \tilde{x}^{(k)}, \tilde{x}^{(l)}) \right] + \widehat{\text{Cov}}(\hat{\xi}_m^{(k)}, \hat{\xi}_r^{(l)}),\end{aligned}$$

where $\tilde{x}^{(k)} = (x_1^{(k)}, \dots, x_N^{(k)})^\top$, and $\hat{\mathbb{E}} \left[\widehat{\text{Cov}}(\xi_m^{(k)}, \xi_r^{(l)} \mid \tilde{x}^{(k)}, \tilde{x}^{(l)}) \right] \neq 0$ in general. This indicates that the sample covariance of the FPC scores predicted by PACE is not a consistent estimator of $\text{Cov}(\xi_m^{(k)}, \xi_r^{(l)})$.

Below, we provide our remedy for this issue. Simple calculations lead to

$$\begin{aligned}\text{Cov}(\xi_m^{(k)}, \xi_r^{(l)}) &= \text{Cov}(\langle X_c^{(k)}(t_k), \phi_m^{(k)}(t_k) \rangle_2, \langle X_c^{(l)}(u_l), \phi_r^{(l)}(u_l) \rangle_2) \\ &= \int_{T_k} \int_{T_l} \text{Cov}(X_c^{(k)}(t_k), X_c^{(l)}(u_l)) \phi_m^{(k)}(t_k) \phi_r^{(l)}(u_l) dt_k du_l.\end{aligned}\quad (3.1)$$

With (3.1), numerical integration approaches could be applied to estimate $\text{Cov}(\xi_m^{(k)}, \xi_r^{(l)})$ after $\text{Cov}(X_c^{(k)}(t_k), X_c^{(l)}(u_l))$ and $\phi_m^{(k)}(t_k)$ being estimated for $1 \leq k \neq l \leq p$ and $m = 1, \dots, M^+$.

Let $\tilde{G}_{kl}(t_{i,j}^{(k)}, t_{i,j'}^{(l)}) = \{x_{i,j}^{(k)} - \hat{\mu}^{(k)}(t_{i,j}^{(k)})\} \{x_{i,j'}^{(l)} - \hat{\mu}^{(l)}(t_{i,j'}^{(l)})\}$ be a raw estimate of cross-covariance surface with $\hat{\mu}^{(k)}(t_k)$ obtained by (2.7). To estimate the cross-covariance function, we again employ a 2-dimensional local linear smoother; specifically, $\hat{G}_{kl}(t_k, u_l) = \hat{\kappa}_0^{(kl)}(t_k, u_l)$ with

$$\begin{aligned}& \left(\hat{\kappa}_0^{(kl)}, \hat{\kappa}_1^{(kl)}, \hat{\kappa}_2^{(kl)} \right) = \\ & \arg \min_{(\kappa_0^{(k)}, \kappa_1^{(k)}, \kappa_2^{(k)})} \sum_{i=1}^N \sum_{\substack{1 \leq j \leq n_i^{(k)} \\ 1 \leq j' \leq n_i^{(l)} \\ t_{i,j}^{(k)} \neq t_{i,j'}^{(l)}}} \left\{ g(\kappa, (t_k, u_l), (t_{i,j}^{(k)}, t_{i,j'}^{(l)})) \right\}^2 K_2 \left(\frac{t_{i,j}^{(k)} - t_k}{h_{G_{kl}}}, \frac{t_{i,j'}^{(l)} - u_l}{h_{G_{kl}}} \right),\end{aligned}\quad (3.2)$$

where $g\left(\boldsymbol{\kappa}, (t_k, u_l), (t_{i,j}^{(k)}, t_{i,j'}^{(l)})\right) = \tilde{G}_{kl}\left(t_{i,j}^{(k)}, t_{i,j'}^{(l)}\right) - \kappa_0 - \kappa_1(t_k - t_{i,j}^{(k)}) - \kappa_2(u_l - t_{i,j'}^{(l)})$.

From (2.5), we know when $k = l$,

$$\text{Cov}\left(\xi_m^{(k)}, \xi_r^{(k)}\right) = \begin{cases} \lambda_m^{(k)} & m = r \\ 0 & m \neq r \end{cases},$$

where $\lambda_m^{(k)}$ can be estimated by eigendecomposition of $\hat{G}_{kk}(t_k, u_k)$.

After estimating $\text{Cov}\left(\xi_m^{(k)}, \xi_r^{(l)}\right)$ for $1 \leq k, l \leq p$, we can reconstruct the matrix $\mathbf{A}$ and denoted as $\mathbf{A}'$ for comparison with the method of [Happ and Greven \(2018\)](#). The estimation of each element $A_{mr}^{[kl]}$ in our method can be represented as

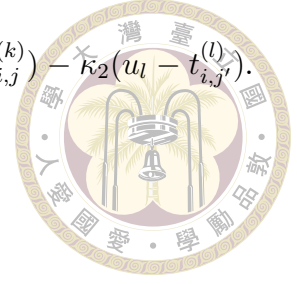
$$A_{mr}^{[kl]} = \int_{\mathcal{T}_k} \int_{\mathcal{T}_l} \widehat{\text{Cov}}\left(X_c^{(k)}(t_k), X_c^{(l)}(u_l)\right) \hat{\phi}_m^{(k)}(t_k) \hat{\phi}_r^{(l)}(u_l) dt_k du_l, \quad (3.3)$$

where $\hat{\phi}_m^{(k)}(t_k)$ and $\hat{\phi}_r^{(l)}(u_l)$ are the estimates from (2.9), and $\widehat{\text{Cov}}\left(X_c^{(k)}(t_k), X_c^{(l)}(u_l)\right)$ estimated by (3.2). Additionally, We can find that if $k = l$, $A_{mr}^{[kl]} = \hat{\lambda}_m^{(k)} I(m = r)$.

3.1 Proposed 1: A Simple Modification from [Happ and Greven \(2018\)](#)

To enable the method of [Happ and Greven \(2018\)](#) to be applicable to different sampling schemes, we replace $\mathbf{A}$ with $\mathbf{A}'$ when applying Proposition 5 in [Happ and Greven \(2018\)](#). With properly estimated (2.16) and (2.17), we can reconstruct the i th multivariate functional data with a chosen truncation lag M as

$$\hat{\mathbf{x}}_i(\mathbf{t}) = \hat{\boldsymbol{\mu}}(\mathbf{t}) + \sum_{m=1}^M \hat{\eta}_{i,m} \hat{\boldsymbol{\psi}}_m(\mathbf{t}),$$



where $\hat{\mathbf{x}}_i(\mathbf{t}) = (\hat{x}_i^{(1)}(t_1), \dots, \hat{x}_i^{(p)}(t_p))$,

$$\begin{aligned}\hat{\psi}_m^{[k]}(t_k) &= \sum_{j=1}^{M^{(k)}} [\mathbf{a}'_m]_j^{[k]} \hat{\phi}_j^{(k)}(t_k), \\ \hat{\eta}_{i,m} &= \sum_{k=1}^p \sum_{j=1}^{M^{(k)}} [\mathbf{a}'_m]_j^{[k]} \hat{\xi}_{i,j}^{(k)},\end{aligned}\tag{3.4}$$

$\hat{\mu}(\mathbf{t})$ is estimated with (2.7), and $\hat{\xi}_{i,j}^{(k)}, \hat{\phi}_j^{(k)}(t_k)$ are calculated by (2.9). $M^{(k)}$ and M are, respectively, the truncation lags corresponding to univariate and multivariate Karhunen-Loève expansions; they can be selected by FVE, i.e.,

$$M^{(k)} = \arg \min_Q \left(\frac{\sum_{m=1}^Q \hat{\lambda}_m^{(k)}}{\sum_{m=1}^{L_k} \hat{\lambda}_m^{(k)}} \geq 1 - q_k \right),\tag{3.6}$$

$$M = \arg \min_Q \left(\frac{\sum_{m=1}^Q \hat{\omega}_m}{\sum_{m=1}^L \hat{\omega}_m} \geq 1 - q \right),\tag{3.7}$$

where $\hat{\lambda}_m^{(k)}$ and $\hat{\omega}_m$ are, respectively, estimated by (2.9) and (2.14), and L_k and L are the non-negative number of $\hat{\lambda}_m^{(k)}$ and $\hat{\omega}_m$, respectively. We chose a certain q_k in our experiments, while directly specified the truncation lag M in order to compare various MFPCA methods fairly.

3.2 Proposed 2: A Weighted Least Squares Approach to Predict the MFPC Scores

Chiou et al. (2014) utilized weighted least squares (WLS) to estimate MFPC scores in the proposed normalized MFPCA method, and claimed that WLS is suitable for both dense and sparse data. Therefore, we also apply WLS to estimate the MFPC scores as a comparison.



Table 3.1: Computational Steps of Proposed 1.



Proposed 1

- Step 1 For $k, l = 1, \dots, p$, obtain $\hat{\mu}^{(k)}(t_k)$ and $\hat{G}_{kl}(t_k, u_l)$ by (2.7) and (2.8), (3.2), respectively.
- Step 2 For $k = 1, \dots, p$, perform FPCA on $\{x_i^{(k)}(t_k)\}_{i=1}^N$ with $M^{(k)}$ selected by FVE (3.6), and $\hat{\lambda}_m^{(k)}, \hat{\phi}_m^{(k)}$, and $\hat{\xi}_{im}^{(k)}, \forall m = 1, \dots, M^{(k)}$ are obtained by (2.9).
- Step 3 Follow (3.3) to calculate $A_{mr}^{[kl]}$, and obtain $\hat{\omega}_m, \mathbf{a}'_m$ by performing eigendecomposition on $\mathbf{A}', \forall m = 1, \dots, M^+$.
- Step 4 Choose M by FVE (3.7), and obtain the estimated MFPCs $\hat{\psi}_m^{[k]}(t_k)$ and MFPC scores $\hat{\eta}_{i,m}, \forall m = 1, \dots, M$, by (3.4) and (3.5), respectively.
- Step 5 Reconstruct data: $\hat{x}_i^{(k)}(t_k) = \hat{\mu}^{(k)}(t_k) + \sum_{m=1}^M \hat{\eta}_{i,m} \hat{\psi}_m^{[k]}(t_k)$.

Define $N_i = \sum_k n_i^{(k)}$, and $\tau_{ki,j}^2 = \sigma_k^2 / V_k(t_{i,j}^{(k)})$. The weights of the WLS method are

$$\Upsilon = \text{Cov}(\varepsilon_i) = \text{diag} \left(\tau_{1i,1}^2, \dots, \tau_{1i,n_i^{(1)}}^2, \dots, \tau_{pi,1}^2, \dots, \tau_{pi,n_i^{(p)}}^2 \right) \in \mathbb{R}^{N_i \times N_i}.$$

Let $\mathbf{z}_i = (z_i^{(1)}, \dots, z_i^{(p)})^\top$ with $z_i^{(k)} = V_k^{-1/2} \left(x_i^{(k)} - \mu^{(k)} \right)$ be the observed normalized data, and $\boldsymbol{\eta}_i = (\eta_{i,1}, \dots, \eta_{i,M})^\top \in \mathbb{R}^M$, where $\eta_{i,m} = \langle \mathbf{z}_i, \psi_m \rangle_{\mathcal{H}}$. Also let $\boldsymbol{\psi}_i = (\psi_{1i}, \dots, \psi_{Mi})^\top \in \mathbb{R}^{M \times N_i}$, where $\psi_{mi} = \left(\psi_{mi}^{[1]\top}, \dots, \psi_{mi}^{[p]\top} \right)^\top \in \mathbb{R}^{N_i}$ and $\psi_{mi}^{[k]} = \left(\psi_m^{[k]}(t_{i,1}), \dots, \psi_m^{[k]}(t_{i,n_i^{(k)}}) \right)^\top \in \mathbb{R}^{n_i^{(k)}}$. The estimated MFPC scores for the i th sample can be expressed as

$$\hat{\boldsymbol{\eta}}_i = \arg \min_{\boldsymbol{\eta}_i} (\mathbf{z}_i - \boldsymbol{\psi}_i^\top \boldsymbol{\eta}_i)^\top \Upsilon^{-1} (\mathbf{z}_i - \boldsymbol{\psi}_i^\top \boldsymbol{\eta}_i) \quad (3.8)$$

$$= (\boldsymbol{\Omega}_i \boldsymbol{\psi}_i^\top)^{-1} \boldsymbol{\Omega}_i \mathbf{z}_i, \quad (3.9)$$

where $\boldsymbol{\Omega}_i = \boldsymbol{\psi}_i \Upsilon^{-1}$. By (3.9), we can simply replace $\boldsymbol{\psi}_i$ and $\tau_{ki,j}^2$ of Υ with their estima-

tors, respectively, to get the estimated MFPC scores, which can be represented as

$$\hat{\eta}_i = (\hat{\eta}_{i,1}, \dots, \hat{\eta}_{i,M})^\top = \left(\hat{\Omega}_i \hat{\psi}_i^\top \right)^{-1} \hat{\Omega}_i \mathbf{z}_i \quad (3.10)$$

with $\hat{\Omega}_i = \hat{\psi}_i \hat{\Upsilon}^{-1}$ and $\hat{\Upsilon} = \text{diag} \left(\hat{\tau}_{1i,1}^2, \dots, \hat{\tau}_{1i,n_i^{(1)}}^2, \dots, \hat{\tau}_{pi,n_i^{(p)}}^2, \dots, \hat{\tau}_{pi,n_i^{(p)}}^2 \right)$.

Table 3.2: Computational Steps of Proposed 2.

Proposed 2

- Step 1 For $k = 1, \dots, p$, obtain $\hat{\mu}^{(k)}(t_k)$ and $\hat{V}_k(t_k)$ by (2.7) and (2.8), respectively, and normalize data $z_i^{(k)}(t_k) = \hat{V}_k^{-1/2}(t_k) \left(x_i^{(k)}(t_k) - \hat{\mu}^{(k)}(t_k) \right)$.
- Step 2 For $k = 1 \dots, p$, perform FPCA on $\{z_i^{(k)}(t_k)\}_{i=1}^N$ with $M^{(k)}$ selected by FVE (3.6), and $\hat{\lambda}_m^{(k)}$, $\hat{\phi}_m^{(k)}$, and $\hat{\xi}_m^{(k)}$, $\forall m = 1, \dots, M^{(k)}$ are obtained by (2.9).
- Step 3 Follow (3.2) and (3.3) to calculate $\widehat{\text{Cov}}(Z^{(k)}(t_k), Z^{(l)}(u_l))$ and $A_{mr}^{[kl]}$, respectively, and obtain $\hat{\omega}_m$, $\mathbf{a}'_m$ by performing eigendecomposition on $\mathbf{A}'$, $\forall m = 1, \dots, M^+$.
- Step 4 Choose M by FVE (3.7), and obtain the estimated MFPCs $\hat{\psi}_m^{[k]}(t_k)$ and MFPC scores $\hat{\eta}_{i,m}$, $\forall m = 1, \dots, M$ by (3.4) and (3.10), respectively.
- Step 5 Reconstruct data: $\hat{x}_i^{(k)}(t_k) = \hat{\mu}^{(k)}(t_k) + \sum_{m=1}^M \hat{\eta}_{i,m} \left(\hat{V}_k^{1/2} \hat{\psi}_m^{[k]} \right) (t_k)$.
-



Chapter 4 Simulation Experiments

In the simulation experiments, we investigate how different sampling schemes and different function complexities among p functions affects the performance of our proposed method compared to the two existing MFPCA methods by [Chiou et al. \(2014\)](#) and [Happ and Greven \(2018\)](#).

4.1 Simulation Setting

The data generation procedure of our simulation experiments is based on that of [Chiou et al. \(2014\)](#). We set the number of variables $p = 3$, and constructed the i th sample as

$$\mathbf{x}_i(\mathbf{t}) = \sum_{m=1}^R \eta_{i,m} \mathbf{D}^{1/2} \psi_m(\mathbf{t}) + \boldsymbol{\varepsilon}_i,$$

$$\text{where } \mathbf{D} = \text{diag}(V_1(t_1), V_2(t_2), V_3(t_3))$$

$$= \text{diag}(t_1 + 0.5, \sin(t_2) + \cos(t_2) + 10, 0.5t_3^2 - t_3 + 1)$$

indicates the variance functions, $\boldsymbol{\varepsilon}_i \stackrel{\text{i.i.d.}}{\sim} N(0, \boldsymbol{\sigma}^2 I)$ with $\boldsymbol{\sigma}^2 = (\sigma_1^2, \sigma_2^2, \sigma_3^2)^\top = (1, 2.25, 0.64)^\top$,

and $R = \min\{60, \#\{\text{positive eigenvalues in simulated data}\}\}$. The MFPC scores $\eta_{i,m}$

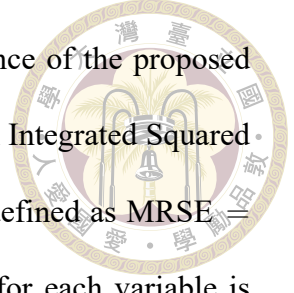
were generated from $N(0, \lambda_m)$. As the procedures of generating $\{\omega_m\}_{m=1}^R$ and $\{\psi_m\}_{m=1}^R$

are complicated, the detail can be found in appendix A.

For simplicity, let the domain of t be $\mathcal{T} = \mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}_3 = [0, 5]$ as [Chiou et al. \(2014\)](#). Through controlling the number of sampling grids and the number of observations for each variable, our data generation procedure yields functional observations under dense, sparse, regular, or irregular data settings. The sampling schemes are summarized in Table 4.1. Notice that when generating dense irregular data, we tripled the number of grids for each variable to create equally spaced pseudo data, and one-third of the grid points are randomly selected with equal probability to obtain the observation time points. In setting 3, since grid numbers are set to vary across each of observations, we first tripled the maximum grid number over the observations for each variable, and then sampled the observed time points. Each sampling scheme was experimented with 3 sample sizes, $N = 100, 200, 500$, and each consisted of $B = 100$ runs. This numerical study was conducted in R with some implementations on the R package `fdapace` from [Carroll et al. \(2020\)](#). To perform FPCA, appropriate bandwidths are required to estimate the mean functions and covariance functions. To ease computational burden, we employed the generalized cross validation (GCV) criterion to select the optimal bandwidths in the first 5 runs each setting, and use the mode of the bandwidths selected in these 5 runs as the bandwidths for the remaining 95 runs.

Table 4.1: Sampling schemes of simulation experiments.

	Data Type	Grid Numbers of $X^{(1)}, X^{(2)}, X^{(3)}$	
		Sampled Data	Predicted Data
Setting 1	dense, regular	25,25,25	40,40,40
Setting 2	dense, irregular	40,40,40	40,40,40
Setting 3	sparse, irregular	5~10,5~10,5~10	40,40,40
Setting 4	dense, regular, grid number varies across variables	37,82,14	50,100,25
Setting 5	dense, irregular, grid number varies across variables	37,82,14	50,100,25



Two criteria are employed to evaluate the numerical performance of the proposed approaches: MRSE (Mean Relative Squared Error) and MISE (Mean Integrated Squared Error). The MRSE is used to evaluate the sample reconstructions, defined as $\text{MRSE} = N^{-1} \sum_i \left(\frac{\|\hat{x}_i - x_i\|_{\mathcal{H}}^2}{\|x_i\|_{\mathcal{H}}^2} \right)$, and the reconstructed path of latent function for each variable is evaluated through MISE, $\text{MISE}_k = N^{-1} \int_{\mathcal{T}_k} \left(\hat{x}_i^{(k)}(t_k) - x_i^{(k)}(t_k) \right)^2 dt_k$. The MISE can further help reveal whether the prediction results of each variable are affected by its functional structure or other settings.

4.2 Simulation Results

In the univariate FPCA step, the number of FPCs are chosen by FVE. As mentioned in (3.6), we used $q_k = 0.005$ for each variable, that is, the first $M^{(k)}$ FPCs explain at least 99.5% variation. Note that [Chiou et al. \(2014\)](#) does not include the univariate FPCA step. While in the MFPCA step, we selected $M = 3$ MFPCs, approximately explaining 90% of the variance in most settings, to compare the results of each method fairly.

Table 4.2 presents the mode of the selected number of FPCs $M^{(1)}, M^{(2)}, M^{(3)}$ under different sample sizes, as well as the average estimated measurement error variance $\hat{\sigma}_1^2, \hat{\sigma}_2^2, \hat{\sigma}_3^2$. It's important to note that since the univariate FPCA procedures for the 3 MFPCA methods (except for [Chiou et al. \(2014\)](#)) are the same, the results have no difference among these methods. We can observe that under dense and regular settings (Setting 1, 4), more FPCs are selected, especially in Setting 4 where the grid numbers for the first two variables are higher, containing more information and thus requiring more FPCs to capture most of the data variance. For sparse data (Setting 3), it requires fewer FPCs to explain similar amount of data variation due to the limited information contained in the

Table 4.2: The chosen FPC number and estimate of measurement error variance of $X^{(1)}$, $X^{(2)}$, $X^{(3)}$ in FPCA for simulation experiments.

	Mode of FPC Number			Estimate of Measurement Error Variance		
	n=100	n=200	n=500	n=100	n=200	n=500
Setting 1	9, 9, 8	9, 9, 8	8, 7, 7	1.00 , 2.26 , 0.64	1.00 , 2.25 , 0.64	1.00 , 2.25 , 0.64
Setting 2	5, 4, 4	4, 4, 5	4, 5, 4	0.96 , 2.21 , 0.66	0.97 , 2.19 , 0.64	0.98 , 2.19 , 0.64
Setting 3	3, 3, 3	4, 4, 3	3, 4, 3	0.99 , 2.71 , 0.93	0.81 , 2.77 , 0.75	0.94 , 2.27 , 0.75
Setting 4	13, 23, 6	12, 23, 6	11, 21, 5	1.00 , 2.24 , 0.64	1.00 , 2.24 , 0.64	1.00 , 2.25 , 0.64
Setting 5	4, 8, 3	5, 5, 3	4, 5, 4	0.94 , 2.21 , 0.77	0.94 , 2.23 , 0.79	0.95 , 2.23 , 0.71

data grid points. As for the estimation of measurement error variance, it can be observed that the estimates are very close to the true ones for dense and regular data, while the estimates for sparse data are not as good.

Table 4.3 indicates that Setting 1 and Setting 3 are more sensitive to the sample sizes compared to the other settings, especially Setting 3, where the average MRSE under each MFPCA method exceeds 0.3 when sample size is 100. However, when the sample size increases to 500, except for the method of Happ and Greven (2018), the average MRSEs of the other methods approach 0.1, and the standard deviation also significantly decreases. This indicates that for sparse cases, the sample size needs to be sufficiently large for each MFPCA method to obtain better estimation. Except for the sparse case (Setting 3), it can be found that Happ and Greven (2018) and Proposed 1 perform similarly and both achieve the best results (regardless of sample size). In the dense and irregular situations (Setting 2, 5), Happ and Greven (2018) has the lowest average MRSE, while Proposed 1 are better in dense and regular situations (Setting 1, 4). It is speculated that these two methods are less affected by the number of grid points and hence perform better in dense data scenarios. In the sparse scenarios (Setting 3), Chiou et al. (2014) and Proposed 2 perform better, with the latter showing superior performance. It is hypothesized that in sparse data situations, fewer grid points and FPCs are needed to explain the data characteristics. Moreover, our

Table 4.3: Mean and standard deviation of MRSE (In parentheses: standard deviation).

		Happ (2018)	Chiou (2014)	Proposed 1	Proposed 2
Setting 1	n=100	0.1821 (0.0039)	0.1843 (0.0049)	0.1738 (0.0036)	0.1740 (0.0043)
	n=200	0.1559 (0.0026)	0.1578 (0.0030)	0.1499 (0.0022)	0.1538 (0.0029)
	n=500	0.1332 (0.0015)	0.1352 (0.0018)	0.1284 (0.0015)	0.1320 (0.0017)
Setting 2	n=100	0.1265 (0.0027)	0.1514 (0.0048)	0.1282 (0.0029)	0.1420 (0.0043)
	n=200	0.1270 (0.0016)	0.1494 (0.0036)	0.1279 (0.0018)	0.1407 (0.0030)
	n=500	0.1149 (<0.001)	0.1267 (0.0017)	0.1161 (0.0010)	0.1256 (0.0017)
Setting 3	n=100	0.3278 (0.0205)	0.3190 (0.0338)	0.3049 (0.0201)	0.3092 (0.0236)
	n=200	0.2956 (0.0169)	0.2660 (0.0234)	0.2603 (0.0159)	0.2694 (0.0240)
	n=500	0.2519 (0.0127)	0.2133 (0.0078)	0.2148 (0.0065)	0.2119 (0.0078)
Setting 4	n=100	0.1308 (0.0018)	0.1616 (0.0029)	0.1249 (0.0015)	0.1209 (0.0017)
	n=200	0.1284 (0.0013)	0.1635 (0.0019)	0.1230 (0.0011)	0.1212 (0.0012)
	n=500	0.1144 (<0.001)	0.1494 (0.0010)	0.1104 (<0.001)	0.1094 (<0.001)
Setting 5	n=100	0.1263 (0.0024)	0.1596 (0.0063)	0.1273 (0.0030)	0.1309 (0.0034)
	n=200	0.1254 (0.0023)	0.1620 (0.0035)	0.1251 (0.0023)	0.1283 (0.0021)
	n=500	0.1103 (0.0010)	0.1516 (0.0019)	0.1103 (0.0010)	0.1125 (0.0010)

method improves the bias in cross-covariance function estimation under sparse conditions, thus achieving optimal results. We can find that at least one of our two proposed methods outperform the existing two methods in any data settings.

Table 4.4 presents the average computation time for each simulation combination, excluding the first 5 experiments where a GCV criterion is employed to select optimal bandwidths. The computation time is generally proportional to the sample size and is significantly related to the population grid points under each setting. Irregular data is generated from equally spaced pseudo data with much more grid points, hence the computation time is generally longer. Comparing the MFPCA methods, method by [Happ and Greven \(2018\)](#) has the lowest average computation time under each setting, while Proposed 2 has the highest. This is because [Happ and Greven \(2018\)](#) primarily involves univariate FPCA and matrix operations, but both [Chiou et al. \(2014\)](#) and our approaches require an additional smoothing step to estimate the cross-covariance functions. Between the two proposed approaches, Proposed 1 utilizes an eigendecomposition to calculate MFPC scores,

Table 4.4: Average simulation time.



		Happ (2018)	Chiou (2014)	Proposed 1	Proposed 2
Setting 1	n=100	0.14	1.26	1.62	1.35
	n=200	0.23	2.62	3.23	2.65
	n=500	0.52	6.97	7.77	7.21
Setting 2	n=100	4.12	15.49	22.71	20.27
	n=200	4.46	26.05	27.25	24.55
	n=500	11.15	51.75	55.2	53.33
Setting 3	n=100	0.28	0.95	1.04	1.41
	n=200	0.43	1.28	1.56	1.76
	n=500	1.04	2.50	3.13	3.36
Setting 4	n=100	0.23	9.50	8.54	6.49
	n=200	0.39	11.05	11.11	10.74
	n=500	0.78	21.40	22.47	19.93
Setting 5	n=100	6.00	59.75	57.74	65.70
	n=200	11.53	79.00	84.66	95.09
	n=500	34.36	140.18	168.31	155.16

while Proposed 2 requires the WLS estimation and data normalization, leading to a higher computational load; consequently, Proposed 2 requires slightly higher computation time.

Figure 4.1 illustrates the MISE of each variable under different sampling schemes, and indicates that only the results of Setting 3 tend to improve with the sample size increases, while the other settings are not significantly affected by sample size. The range of MISE varies across variables due to the existence of differences in the functional structure of each variable and different settings of measurement error variance. Specifically, $MISE_2$ is consistently larger, likely because $V_2(t_2)$ and σ_2^2 are both larger than the other two variables, potentially requiring more grid points to comprehensively represent the variance of X_2 . Additionally, within the first three settings, Proposed 2 has the lowest $MISE_1$ and $MISE_3$, and the method by Chiou et al. (2014) performs similarly. As for $MISE_2$, the method by Happ and Greven (2018) performs the best in Setting 2, and Proposed 1 performs the best in Settings 1 and 3, $X^{(2)}$ may need more grid points to fully explain its larger variance function. For Setting 4 and 5, where the grid number varies across vari-

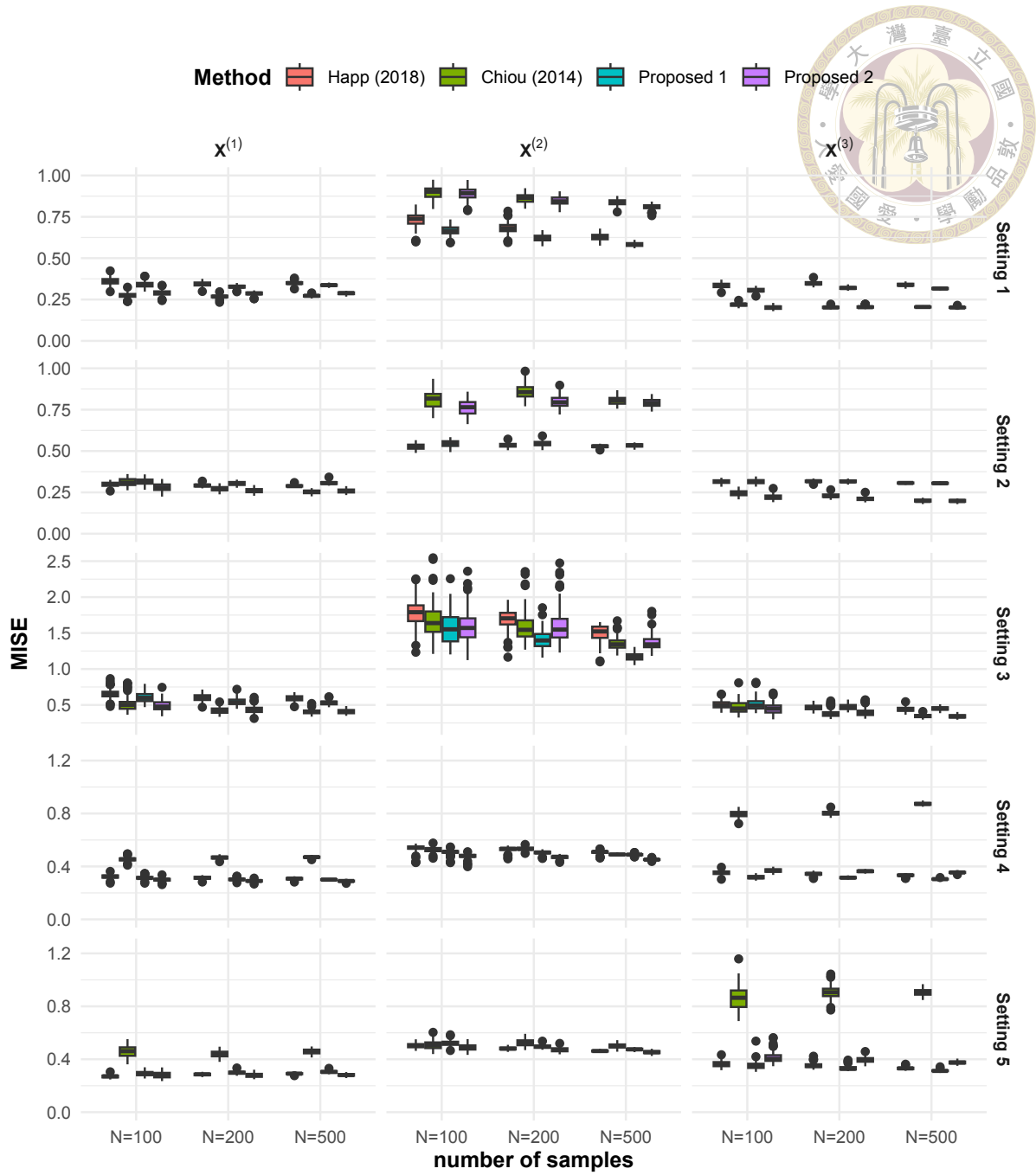


Figure 4.1: $MISE_k$ boxplot of each variable for each sample scheme.

ables, Proposed 2 has the smallest $MISE_1$ and $MISE_2$, while Proposed 1 has the smallest $MISE_3$, with the method by Chiou et al. (2014) performing the worst. In general, one of our proposed methods yields the lowest MISE for almost all variables. Whether the grid points for each variable are the same or not, Proposed 1 performs better than Proposed 2, which is more sensitive to the number of grid points. But if data is sparse, it is more appropriate to use Proposed 2.



Chapter 5 Real Data Analysis

In this chapter, our new proposed methods are applied to medical data from the longitudinal retrospective study ([Mondal, 2023](#)). The main objective of this study is to assess the impact of body mass index (BMI) on the estimated airway resistance (impedance encountered by airflow during respiration) and reactance (elasticity and stiffness of the airway walls) measured by impulse oscillometry (IOS), which is a non-invasive method for assessing respiratory function, in children with sickle cell disease.

The dataset includes the personal information of the study subjects, health data indicators, disease and medication history, and IOS estimates. We take the 4 IOS estimates and BMI from this data for analysis. Let $X^{(1)}$, $X^{(2)}$, $X^{(3)}$, $X^{(4)}$ and $X^{(5)}$ be R5Hz_PP, R20Hz_PP, X5Hz_PP, Fres_PP and the BMI index, respectively. R5Hz_PP and R20Hz_PP are the airway resistance at a frequency of 5Hz and 20Hz, respectively. X5Hz_PP is the airway reactance at a frequency of 5Hz. Fres_PP is a measure of the point at which the reactance of the airways transitions from positive to negative during respiration. The BMI index is the ratio of weight to the square of height. All these variables are time-varying, and the time of interest is the subject's age (recorded in months). Each observation is the multivariate data of subject at ages ranging from 50 to 240 months. The data contains $N = 90$ subjects, each was observed 1 to 5 times at different time points.

Figure 5.1 displays all the observations of the 5 variables analyzed in this study. Each line represents the observations of one subject. Each variable exhibits its own time-varying trend. Each subject has the same observed time points across all variables, except $X^{(4)}$, where a few time points are not observed.

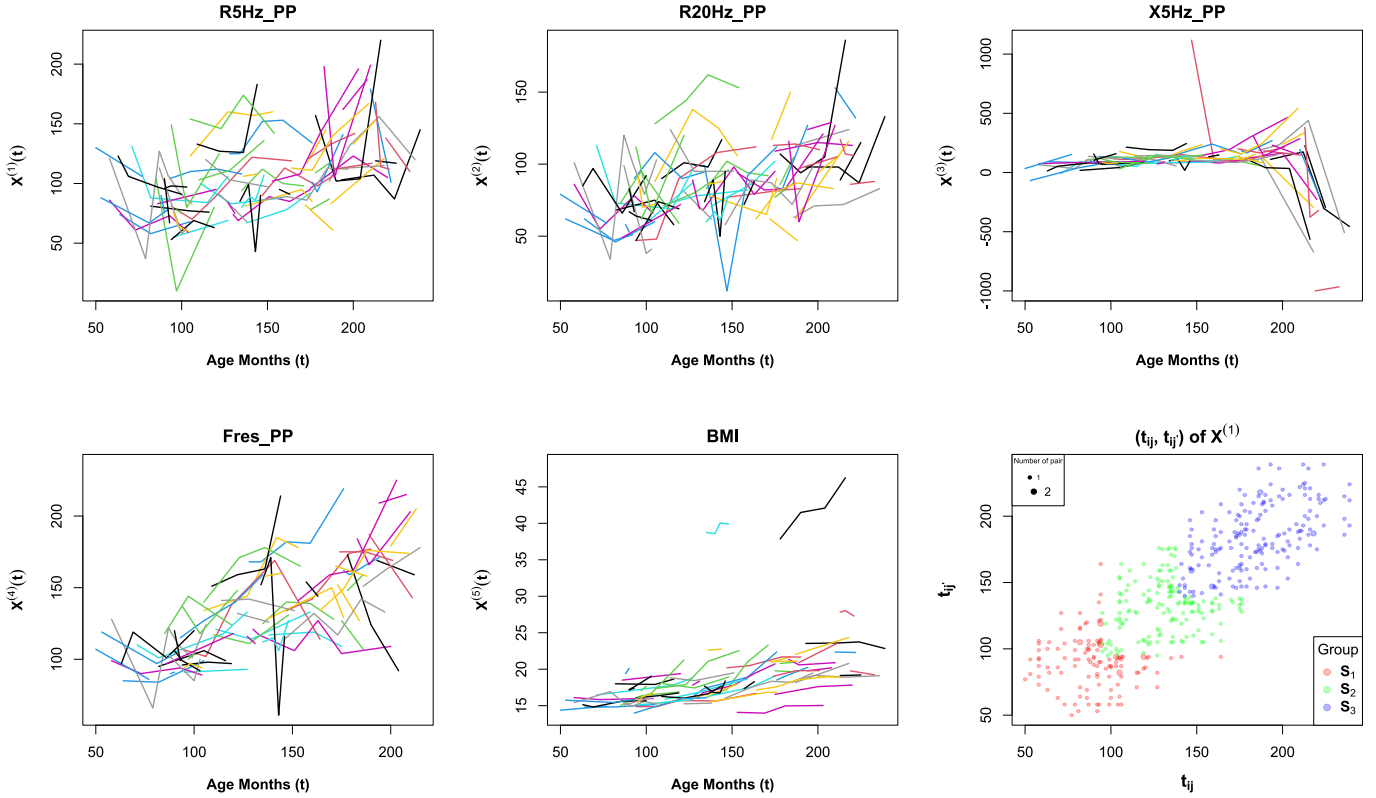


Figure 5.1: The 5 variables of the target dataset (the subplots excluding bottom-right) and the assembled pairs (t_{ij}, t_{ij}') of all subjects (bottom-right subplot).

However, the diagonal pattern observed in the bottom-right subplot of Figure 5.1 indicates that the range of observation time of each subject does not cover the entire time interval. The average observation time span of one subject is only 20 months. This leads to difficulties in estimating the covariance function as indicated in the bottom-right plot of Figure 5.1. To address this issue, we divided the samples into 3 groups by the first observation time point t_{i1} of each subject. The groups are respectively $S_1 = \{i : t_{i1} < 95\}$, $S_2 = \{i : 95 \leq t_{i1} < 140\}$, $S_3 = \{i : 140 \leq t_{i1}\}$.

Due to that the trends (empirical mean functions) are different among groups, we first estimate the overall mean function $\hat{\mu}^{(k)}(t_k)$, $\forall k$ and center the data $x_i'^{(k)}(t_k) = x_i^{(k)}(t_k) - \hat{\mu}^{(k)}(t_k)$, $\forall i, k$. In the second step, the observation time points for each sample are reset by subtracting the first observation time, i.e., $t'_{ij} = t_{ij} - t_{i1}$; $t'_{i1} = 0$. We then estimate the MFPCs and MFPC scores for each group, and finally add $\hat{\mu}^{(k)}(t_k)$ back to obtain the desired $\hat{x}(t)$.

Table 5.1 presents the number of FPCs selected for each variable of the three groups, as well as the estimated measurement error variance. We selected the number of FPCs using $FVE = 0.99$ (the same as $q_k = 0.01$ in (3.6)) and the number of MFPCs using $FVE = 0.9$ (the same as $q = 0.1$ in (3.7)). Two to four FPCs are selected in each group for original data, while one to three FPCs are selected in each group for normalized data. There exist significant differences in the estimate of measurement error variance across the variables. BMI, as a natural indicator of human growth, has almost no measurement error, while the other variables, being manually measured indicators, have relatively larger measurement errors. Additionally, the measurement errors of the same variable vary with groups, and the variables in S_3 generally have higher measurement error variances.

Table 5.1: The chosen FPC number and estimate of measurement error variance in FPCA.

	uFPC Number (Original Data)			uFPC Number (Normalized Data)			Estimate of Measurement Error Variance		
	S_1	S_2	S_3	S_1	S_2	S_3	S_1	S_2	S_3
R5Hz_PP ($X^{(1)}$)	2	2	2	3	2	2	305.08	234.32	939.51
R20Hz_PP ($X^{(2)}$)	4	3	2	3	3	2	265.59	276.01	461.83
X5Hz_PP ($X^{(3)}$)	3	2	3	1	2	3	1181.14	435.80	39061.37
Fres_PP ($X^{(4)}$)	3	3	3	2	1	3	250.19	184.76	443.23
BMI ($X^{(5)}$)	2	2	3	2	3	3	0.88	1E-06	1E-06

Define NMSE (normalized mean squared error) and NMSE_k , $k = 1, \dots, 5$ as following:

$$\text{NMSE} = N^{-1} \sum_{i=1}^N \left[\sum_{k=1}^p \sum_{j=1}^{n_i^{(k)}} \left(\hat{x}_i^{(k)}(t_{i,j}) - x_i^{(k)}(t_{i,j}) \right)^2 / \sum_{k=1}^p \sum_{j=1}^{n_i^{(k)}} \left(x_i^{(k)}(t_{i,j}) \right)^2 \right],$$

$$\text{NMSE}_k = N^{-1} \sum_{i=1}^N \left[\sum_{j=1}^{n_i^{(k)}} \left(\hat{x}_i^{(k)}(t_{i,j}) - x_i^{(k)}(t_{i,j}) \right)^2 / \sum_{j=1}^{n_i^{(k)}} \left(x_i^{(k)}(t_{i,j}) \right)^2 \right].$$

Table 5.2 summarizes the NMSE and NMSE_k of all 4 methods and indicates Proposed 2 achieves the lowest NMSE, while Proposed 1 falls between the existing two methods. In the simulation experiments, we have observed that as the data becomes denser, Proposed 1 shows more advantage, whereas Proposed 2 performs better when the data is sparser or when the grid number is large. The results in Table 5.2 are consistent with our perceptions in the simulation experiments. Additionally, Proposed 2 has the best NMSE_k across all variables. Comparing the other three methods, except for NMSE_5 , [Happ and Greven \(2018\)](#) and Proposed 1 perform similarly. However, [Chiou et al. \(2014\)](#) shows a better NMSE_5 , corresponding to BMI, which has a smaller measurement error and variance function. This is also consistent with the inference from the simulation experiments, where [Chiou et al. \(2014\)](#) and Proposed 2 were found to be better when the variance function is relatively small or the grid number is relatively large.

Table 5.2: NMSE and NMSE_k (for each variable) of the 4 compared methods.

	Happ (2018)	Chiou (2014)	Proposed 1	Proposed 2
NMSE	0.0817	0.1255	0.0853	0.0566
NMSE₁	0.0247	0.0637	0.0294	0.0134
NMSE₂	0.0287	0.0517	0.0340	0.0186
NMSE₃	0.1786	0.2629	0.1763	0.1504
NMSE₄	0.0118	0.0294	0.0106	0.0046
NMSE₅	0.0191	0.0185	0.0287	0.0236

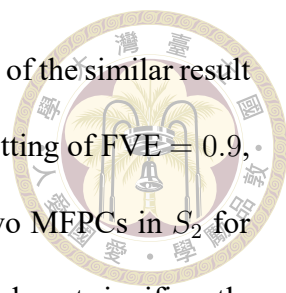


Figure 5.2 presents the MFPCs estimated in Proposed 2. Because of the similar result in Proposed 1, only the results of Proposed 2 are shown. Under the setting of $FVE = 0.9$, S_1 , S_2 , and S_3 selected 4, 3, and 5 MFPCs respectively. The first two MFPCs in S_2 for each variable are nearly horizontal lines, indicating that these MFPCs do not significantly vary over time after removing the mean function. In S_1 , with t increasing, $\Psi_1^{[k]}(t_k)$ for each variable (except for BMI) shows an initial flattening followed by a slowly decreasing trend. For $\Psi_2^{[k]}(t_k)$, variables related to airway resistance (R5Hz_PP, R20Hz_PP) and airway reactance (X5Hz_PP) exhibit strong negative correlations. This observation aligns with existing literature suggesting a stronger correlation between airway resistance and reactance in younger individuals ([Haruna et al., 2010](#); [Schumm et al., 2024](#)). Higher airway resistance or lower reactance indicates poorer airway function, which may correlate with respiratory diseases. Overall, the estimated MFPCs in S_2 show the least variation over time across all variables. In contrast, the estimated MFPCs in S_1 exhibit a more noticeable trend of change over time across all variables, while the estimated MFPCs in S_3 exhibit more pronounced fluctuations.

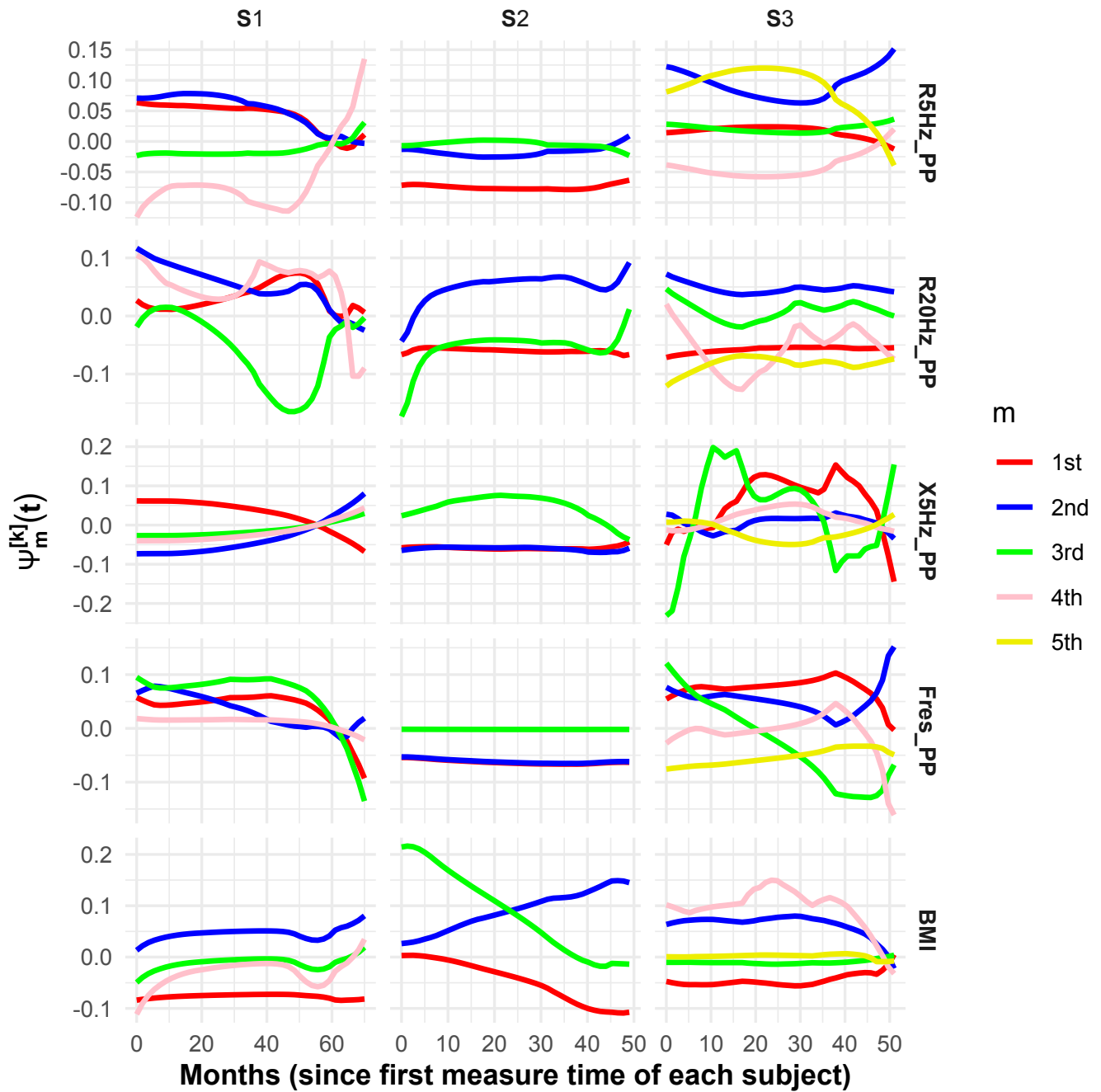


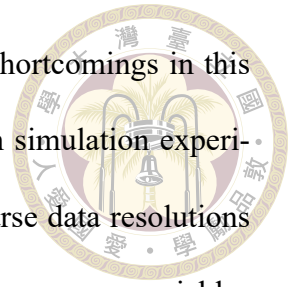
Figure 5.2: Estimated MFPCs of Proposed 2.



Chapter 6 Conclusion and Discussion

To address the estimation issues encountered with sparse data, we modified the elements in the matrix linking FPCA and MFPCA in [Happ and Greven \(2018\)](#) and combined it with the normalization procedure and WLS estimation for MFPC scores in [Chiou et al. \(2014\)](#). We proposed two hybrid MFPCA approaches. Our numerical experiments show that [Chiou et al. \(2014\)](#) is more sensitive to data resolution, but performs better when the resolution is sufficiently high. For dense data, the method that only modifies the matrix in [Happ and Greven \(2018\)](#) (Proposed 1) shows superior performance. Given that [Happ and Greven \(2018\)](#) is suitable for various resolutions, Proposed 1 is less sensitive to different resolution settings than Proposed 2. However, for sparse data, normalizing the data with the MFPC scores estimated by WLS (Proposed 2) yields significantly better results. It is speculated that in the case of sparse data, the weights constructed with measurement error variance and variance function result in less biased MFPC score calculations, and data normalization also leads to a more robust estimate of covariance function. In the real data analysis, due to inherent limitations, we divided the data into three groups based on the first observation time of the samples, retaining only the relative time information for each subject in the analysis of each group. For this sparse data, whether considering the entire dataset or individual variables, Proposed 2 consistently performs the best. This demonstrates that our proposed method effectively addresses the challenges previ-

ously encountered in sparse MFPCA studies. There are still some shortcomings in this study. For example, we considered only one case for sparse data in simulation experiments. It may be necessary to investigate other combinations of sparse data resolutions or to use measurement error variance, variance function, and domain ranges as variables to further explore their impact on MFPCA estimation. Additionally, although this study aims to apply the method to multidimensional data, due to time and capability constraints, multidimensional scenarios were not considered in the simulation experiments and real data analysis, and different resolutions for each variable were set as a temporary substitute. Future researchers interested in this topic might prioritize attempting the analysis of multidimensional data.





References

- Abrahamsen, P. et al. (1997). A review of Gaussian random fields and correlation functions. Norsk Regnesentral/Norwegian Computing Center.
- Abramowitz, M. and Stegun, I. A. (1968). Handbook of mathematical functions with formulas, graphs, and mathematical tables, volume 55. US Government printing office.
- Allen, G. I. (2013). Multi-way functional principal components analysis. In 2013 5th IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), pages 220–223.
- Berrendero, J. R., Justel, A., and Svarc, M. (2011). Principal components for multivariate functional data. Computational Statistics & Data Analysis, 55(9):2619–2634.
- Carroll, C., Gajardo, A., Chen, Y., Dai, X., Fan, J., Hadjipantelis, P. Z., Han, K., Ji, H., Mueller, H., and Wang, J.-L. (2020). fdapace: Functional data analysis and empirical dynamics. R package version 0.5.
- Chen, L.-H. and Jiang, C.-R. (2017). Multi-dimensional functional principal component analysis. Statistics and Computing, 27:1181–1192.
- Chen, M. G. (2023). Estimating sparsely and irregularly observed multivariate functional data. The ITEA Journal of Test and Evaluation, 44(3).

Chen, Y., Härdle, W. K., He, Q., and Majer, P. (2015). Risk related brain regions detected with 3d image fpca. SSRN Electronic Journal.

Chiou, J.-M., Chen, Y.-T., and Yang, Y.-F. (2014). Multivariate functional principal component analysis: A normalization approach. Statistica Sinica, 24(4):1571–1596.

Consagra, W., Venkataraman, A., and Qiu, X. (2024). Efficient multidimensional functional data analysis using marginal product basis systems. Journal of Computational and Graphical Statistics, 33(2):567–577.

Dai, W. and Genton, M. G. (2018). An outlyingness matrix for multivariate functional data classification. Statistica Sinica, 28(4):2435–2454.

Di Salvo, F., Ruggieri, M., and Plaia, A. (2015). Functional principal component analysis for multivariate multidimensional environmental data. Environmental and Ecological Statistics, 22:739–757.

Happ, C. and Greven, S. (2018). Multivariate functional principal component analysis for data observed on different (dimensional) domains. Journal of the American Statistical Association, 113(522):649–659.

Haruna, A., Oga, T., Muro, S., Ohara, T., Sato, S., Marumo, S., Kinose, D., Terada, K., Nishioka, M., Ogawa, E., et al. (2010). Relationship between peripheral airway function and patient-reported outcomes in copd: a cross-sectional study. BMC Pulmonary Medicine, 10:1–7.

Huang, J. Z., Shen, H., and Buja, A. (2009). The analysis of two-way functional data using two-way regularized singular value decompositions. Journal of the American Statistical Association, 104(488):1609–1620.



Jacques, J. and Preda, C. (2014). Model-based clustering for multivariate functional data. Computational Statistics & Data Analysis, 71:92–106.

James, G. M., Hastie, T. J., and Sugar, C. A. (2000). Principal component models for sparse functional data. Biometrika, 87(3):587–602.

Kayano, M. and Konishi, S. (2009). Functional principal component analysis via regularized gaussian basis expansions and its application to unbalanced data. Journal of Statistical Planning and Inference, 139(7):2388–2398.

Kokoszka, P. (2012). Inference for Functional Data with Applications. Springer New York.

Li, C., Xiao, L., and Luo, S. (2020). Fast covariance estimation for multivariate sparse functional data. Stat, 9(1):e245.

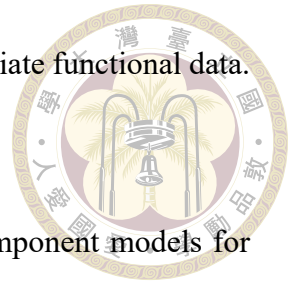
Li, Y., Huang, C., and Härdle, W. K. (2019). Spatial functional principal component analysis with applications to brain image data. Journal of Multivariate Analysis, 170:263–274.

Mercer, J. (1909). Xvi. functions of positive and negative type, and their connection the theory of integral equations. Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of A Mathematical or Physical Character, 209:415–446.

Minasny, B. and McBratney, A. B. (2005). The matern function as a general model for soil variograms. Geoderma, 128(3-4):192–207.

Mondal, P. (2023). Impact of BMI on IOS measures. <http://dx.doi.org/10.17632/rwcvfz9np9.1>.

Ramsay, J. O. (1982). When the data are functions. Psychometrika, 47:379–396.



Ramsay, J. O. and Silverman, B. W. (2005). Functional Data Analysis. Springer.

Rice, J. A. and Wu, C. O. (2001). Nonparametric mixed effects models for unequally sampled noisy curves. Biometrics, 57(1):253–259.

Schumm, B., Bremer, S., Knödlseeder, K., Schönfelder, M., Hain, R., Semmler, L., Lorenz, E., Wackerhage, H., Kähler, C. J., and Jörres, R. (2024). Indices of airway resistance and reactance from impulse oscillometry correlate with aerosol particle emission in different age groups. Scientific Reports, 14(1):4644.

Shen, S., Zhou, H., He, K., and Zhou, L. (2023). Principal component analysis of two-dimensional functional data with serial correlation. Journal of Agricultural, Biological and Environmental Statistics.

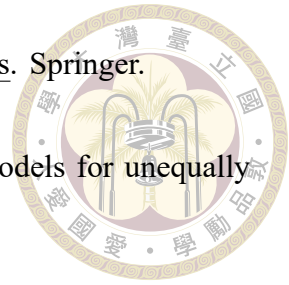
Shi, H., Jiang, S., Ma, D., Beg, M. F., and Cao, J. (2024). Dynamic survival prediction using sparse longitudinal images via multi-dimensional functional principal component analysis. Journal of Computational and Graphical Statistics, pages 1–12.

Song, J. and Li, B. (2021). Nonlinear and additive principal component analysis for functional data. Journal of Multivariate Analysis, 181:104–675.

Staniswalis, J. G. and Lee, J. J. (1998). Nonparametric regression analysis of longitudinal data. Journal of the American Statistical Association, 93(444):1403–1418.

Wang, J.-L., Chiou, J.-M., and Müller, H.-G. (2016). Functional data analysis. Annual Review of Statistics and Its Application, 3(1):257–295.

Yao, F., Müller, H.-G., and Wang, J.-L. (2005). Functional data analysis for sparse longitudinal data. Journal of the American Statistical Association, 100(470):577–590.





Appendix A — More Details about Simulation Settings

In chapter 4, we mentioned that the generation procedures of eigenfunctions $\{\psi_m\}$ and their corresponding eigenvalues $\{\lambda_m\}$ in the simulation experiments was relatively complicated. In the following contents, we will provide a detailed description. We mainly construct a multivariate correlation function, using the results of its eigendecomposition as the simulated eigenfunctions and eigenvalues. We consider the following 3 types of correlation functions.

- The Bessel correlation function of the first kind ([Abramowitz and Stegun, 1968](#)),

$$J_\nu(z) = \left(\frac{z}{2}\right)^\nu \sum_{j=0}^{\infty} \frac{(-z^2/4)^j}{j! \Gamma(\nu + j + 1)},$$

with order $\nu = 0$, where $z = |t|/\rho_1^0$.

- The Matérn correlation function ([Minasny and McBratney, 2005](#)),

$$F(z) = \frac{1}{2^{\nu-1}\Gamma(\nu)} z^\nu K_\nu(z),$$



with order $\nu = 2.5$, where $z = \sqrt{2\nu}|t|/\rho_2^o$, where

$$K_\nu(z) = \frac{\pi}{2} \frac{I_{-\nu}(z) - I_\nu(z)}{\sin(\nu\pi)}$$

with the modified Bessel function

$$I_{\pm\nu}(z) = \left(\frac{z}{2}\right)^{\pm\nu} \sum_{j=0}^{\infty} \frac{(z^2/4)^j}{j! \Gamma(j \pm \nu + 1)}.$$

- The rational quadratic correlation function ([Abrahamsen et al., 1997](#))

$$R(z) = \frac{1}{1 + z^2}$$

with $z = t/\rho_3^o$.

Here, the constants ρ_1^o, ρ_2^o and ρ_3^o are scale parameters of the correlation functions, and we set $\rho^o = (1, 2.2, 3)$. We take the $\{1, 2, \dots, N - j + 1\}$ th elements of $J_\nu(z), F(z)$ and $R(z)$ as the elements of the j th row in the upper triangular matrix of the correlation function C_{kk} , where $C_{kk} = C_{kk}^\top$ and N is the number of recording times and $j = 1, \dots, N$. We obtain the eigenfunctions $\{\psi_m\}$ by the following steps.

1. Set the correlation function $C_{kk}(t_k, u_k), k = 1, \dots, 3$, based on the Bessel correlation function, the Matérn correlation function and the rational quadratic correlation function described above. Obtain $\{\lambda_{km}\}$ and $\{\phi_{km}\}$ through the spectral decomposition of C_{kk} such that $C_{kk}(t_k, u_k) = \sum_{m=1}^{\infty} \lambda_{km} \phi_{km}(t_k) \phi_{km}(u_k)$.
2. Construct the cross-correlation functions $C_{kl}(t_k, u_l), k \neq l$, such that $C_{kl}(t_k, u_l) = \sum_{m=1}^{\infty} \bar{\lambda}_m \phi_{km}^*(t_k) \phi_{km}^*(u_l)$, where $\phi_{km}^*(t_k) = \phi_{km}(t_k)/\sqrt{3}$ and $\bar{\lambda}_m = \sum_{k=1}^3 \lambda_{km}$.

3. Based on the spectral decomposition of $C = \{C_{kl}; 1 \leq k, l \leq 3\}$, obtain the eigenfunctions $\{\psi_m\}$ with the corresponding eigenvalues $\{\omega_m\}$ for $m = 1, \dots, M$, where M is the number of positive eigenvalues.

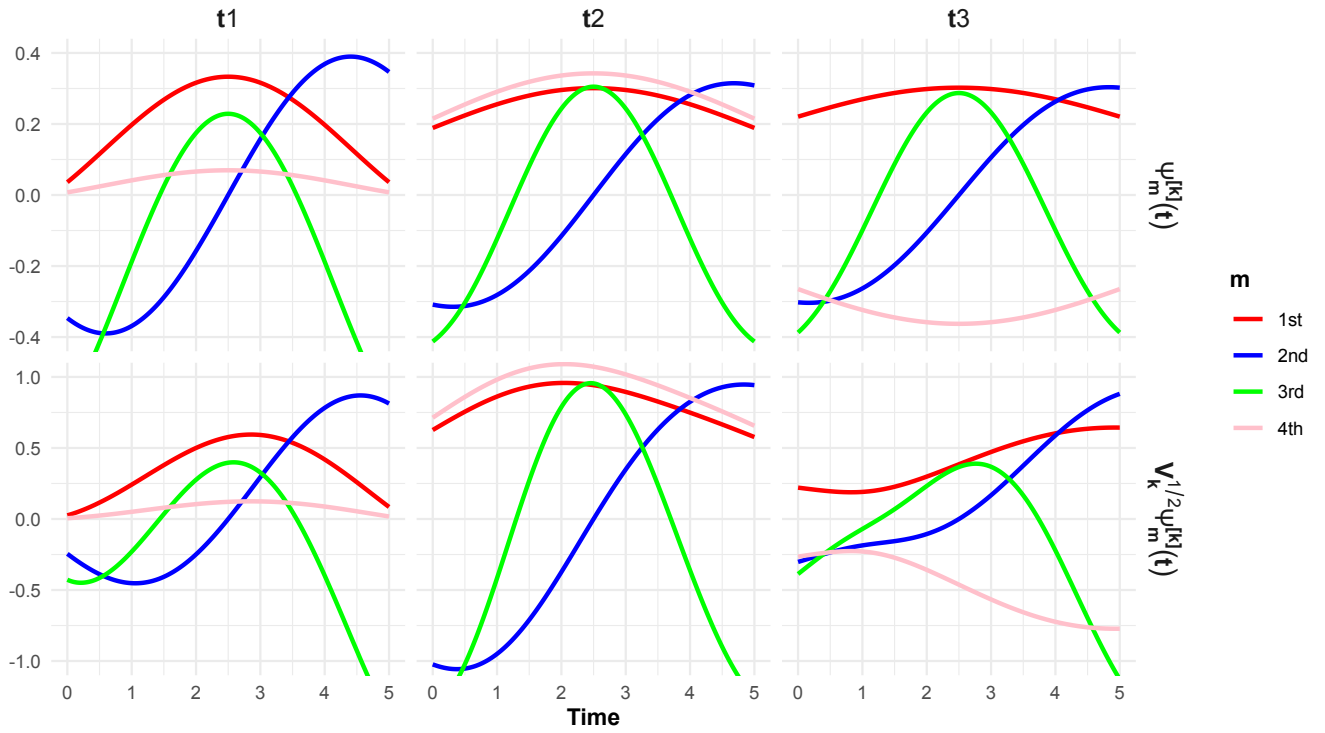
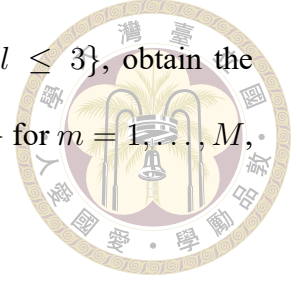
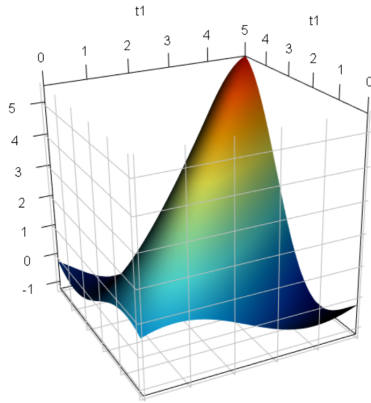
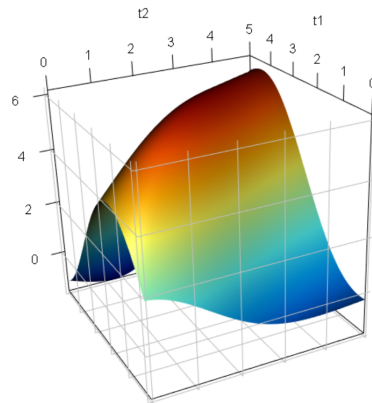


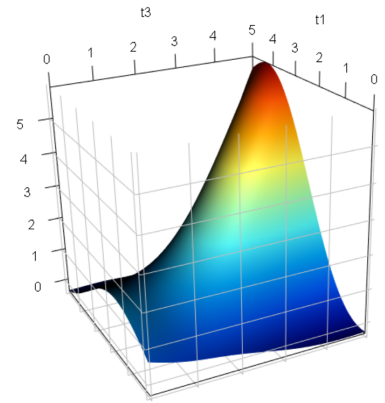
Figure A.1: The first four true MFPCs $\{\Psi_m^{[k]}\}$ in simulation experiments.
First row: MFPCs; second row: MFPCs multiply by the square root of variance functions.



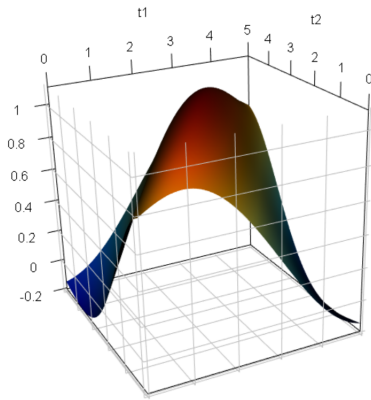
(a) X_1



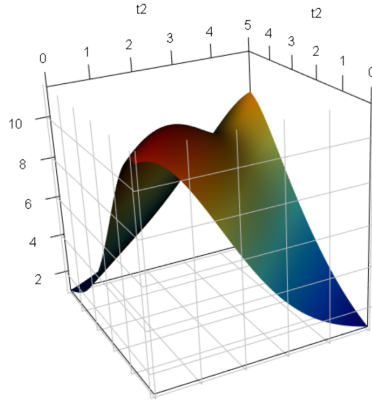
(b) (X_2, X_1)



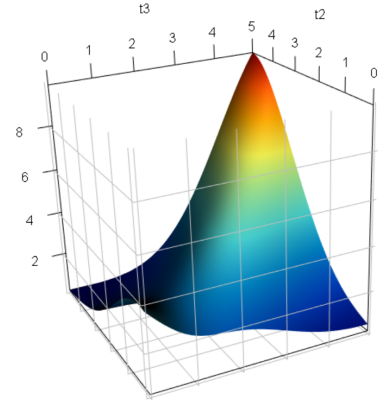
(c) (X_3, X_1)



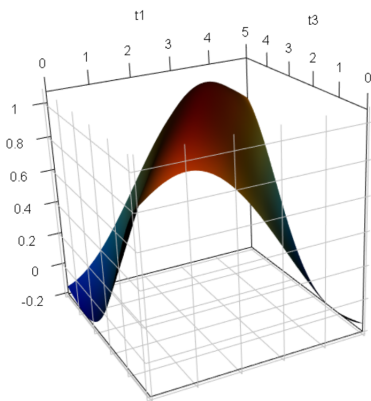
(d) (X_1, X_2)



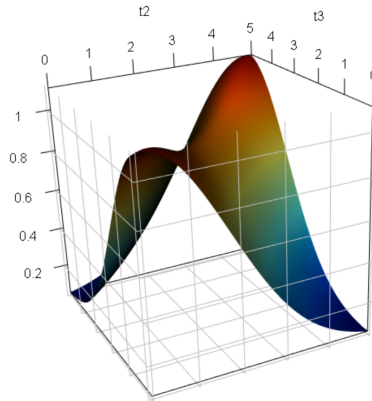
(e) X_2



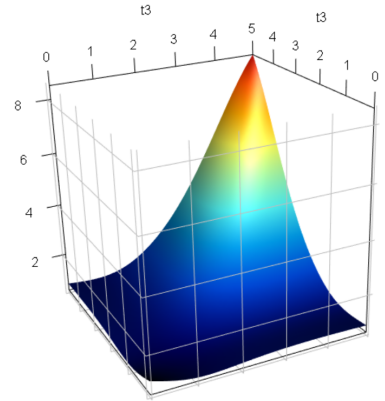
(f) (X_3, X_2)



(g) (X_1, X_3)



(h) (X_2, X_3)



(i) X_3

Figure A.2: The true covariance functions $G_{kk}(t_k, u_k)$ (diagonal part), cross-covariance functions $G_{kl}(t_k, u_l)$ (upper triangular part) and correlation functions $C_{kl}(t_k, u_l)$ (lower triangular part) in simulation experiments.



Appendix B — More Results about Real Data Analysis

Table B.1: Computational time of the 4 compared methods in real data analysis.

	Happ (2018)	Chiou (2014)	Proposed 1	Proposed 2
Time	33.43	56.28	101.77	120.30

Table B.2: Cumulative FVE and the chosen MFPC number (in parentheses) of the 4 compared methods in real data analysis.

	Happ (2018)	Chiou (2014)	Proposed 1	Proposed 2
S_1	0.9550 (3)	0.9345 (6)	0.9447 (5)	0.9417 (4)
S_2	0.9180 (2)	0.9059 (4)	0.9197 (4)	0.9271 (3)
S_3	0.9113 (3)	0.9219 (5)	0.9029 (3)	0.9537 (5)

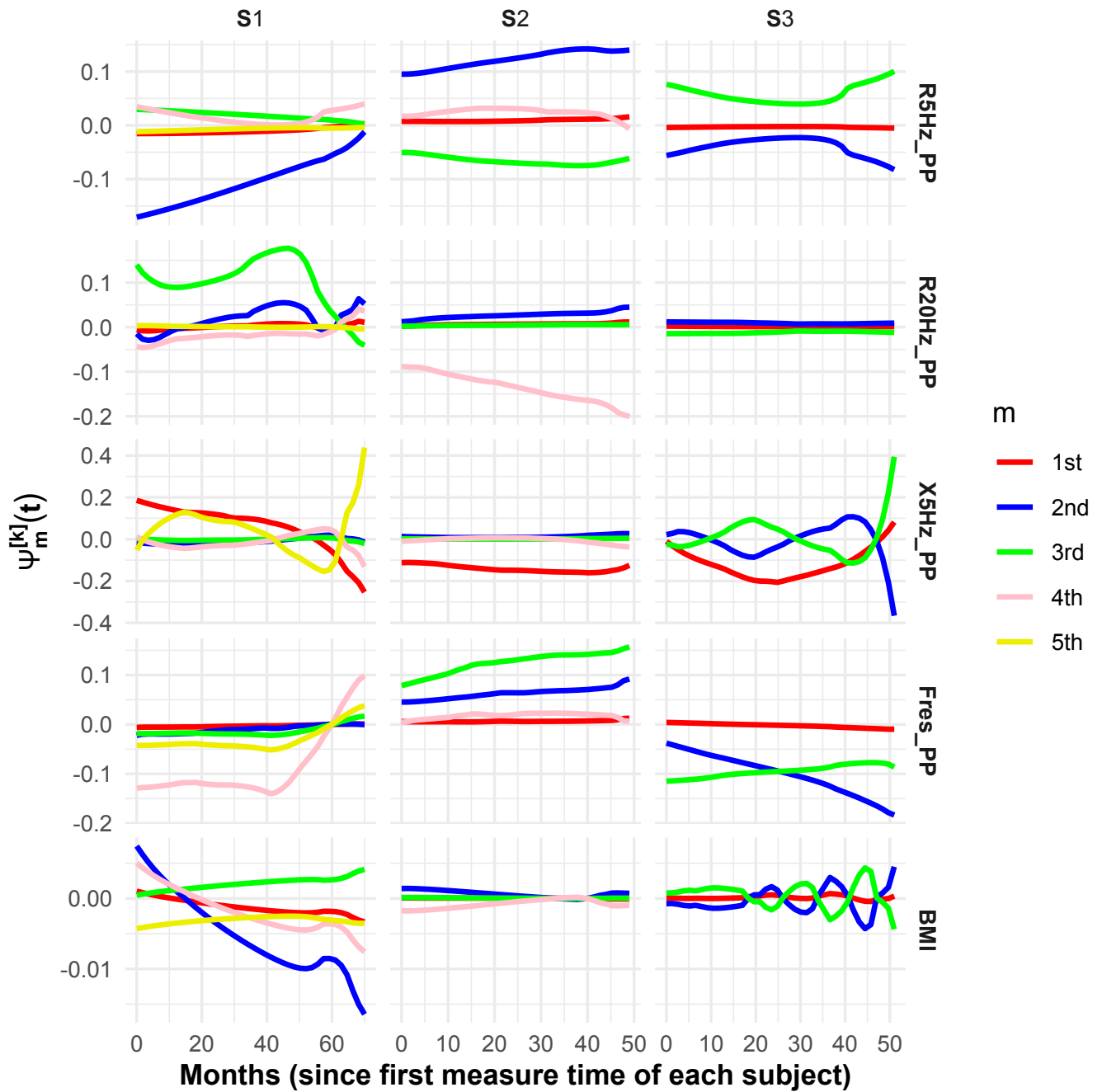


Figure B.3: Estimated MFPCs of Proposed 1 in real data analysis.